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Beyond the Benefits: A Systematic Review of the Harms and Consequences of Generative AI in Computing Education

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Abstract

Generative artificial intelligence (GenAI) has already had a big impact on computing education with prior research identifying many benefits. However, recent studies have also identified potential risks and harms. To continue maximizing AI benefits while addressing the harms and unintended consequences, we conducted a systematic literature review of research focusing on the risks, harms, and unintended consequences of GenAI in computing education. Our search of ACM DL, IEEE Xplore, and Scopus (2022-2025) resulted in 1,677 papers, which were then filtered to 224 based on our inclusion and exclusion criteria. Guided by best practices for systematic reviews, four reviewers independently extracted publication year, learner population, research method, contribution type, GenAI technology, and educational task information from each paper. We then coded each paper for concrete harm categories such as academic integrity, cognitive effects, and trust issues. Our analysis shows patterns in how and where harms appear, highlights methodological gaps and opportunities for more rigorous evidence, and identifies under-explored harms and student populations. By synthesizing these insights, we intend to equip educators, computing students, researchers, and developers with a clear picture of the harms associated with GenAI in computing education.

CCS Concepts

• **Social and professional topics → Computing education**.

Keywords

large language models, generative AI, harms, computing education

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1 Introduction

Over the last few years, generative AI (GenAI) has had a transformative impact on computing education [183, 184]. For example, GenAI can explain code to students better than their peers [122, 138], can provide hints to help students get unstuck [115, 204], and can make learning materials more personally or culturally relevant [30]. Classes have adopted AI-first approaches that introduce the most engaging parts of the computing curriculum earlier and delay more tedious or frustrating aspects of early programming, such as syntax and boilerplate code [181, 200, 243]. Tools are also being developed to more carefully scaffold the use of generative AI in classrooms [53, 54, 112, 126, 137]. For example, Denny et al. introduced Prompt Problems as a technique designed to help students practice prompt engineering by automatically generating and evaluating the resulting code [53]. MacNeil et al. introduced Autocompletion Quizzes to give students practice differentiating between good and bad code suggestions [137]. These two examples show how pedagogical tools can not only facilitate learning, but actually teach new skills and competencies that students need to use AI effectively.

Alongside these benefits and innovative tools and pedagogies, researchers have also raised concerns about negative impacts on learning and unintended consequences. Early concerns included students over-relying on GenAI [29, 119, 254], academic dishonesty [56, 80, 94, 217, 254], and uneven benefits, with the best students receiving more value from AI [95, 187, 254]. Some have even compared this to the 'digital divide' of the 1990s [254]. Subsequent empirical studies have provided evidence that support some these concerns. For example, GenAI models can solve problems at the CS1 and CS2 levels [69, 217], Parsons problems [94, 199], and graph and tree data structure tasks using only image-based inputs [80]. Recent work has also provided empirical evidence for a widening gap between students who can use AI effectively and those who struggle [142, 187]. A recent study by Hou et al. also showed negative impacts on peer interactions and learning communities [93].

These negative impacts are concerning, but they are not yet well-understood. To address this gap, we conducted a systematic literature review of studies on GenAI in computing education that discuss harms, challenges, and unintended consequences. Our review focuses on research published between 2022 and 2025. In this review, we define 'harms' broadly to include cognitive impacts

(e.g., over-reliance), ethical risks (e.g., plagiarism), social and equity concerns, and unintended pedagogical effects. Our goal was to investigate the following research questions:

RQ1: What types of harms or unintended consequences of generative AI have been reported? **(Types of Harms)**
RQ2: What empirical evidence supports each identified harm? **(Evidence)**
RQ3: In what educational contexts have these harms been examined or discussed? **(Contexts)**

Our search resulted in 1,849 references from the three popular computer science databases: ACM Digital Library, IEEE Xplore, and Scopus. After removing duplicate and retracted papers, 1,677 references remained. After filtering these papers based on our inclusion and exclusion criteria, we had 224 total references which were analyzed in detail. We evaluated the types of harms and their prevalence across various contexts including GenAI tool types and populations. We also coded for research contributions, methods, and levels of evidence.

Our findings show a few clear takeaways: (1) researchers are increasingly attending to the potential harms of generative AI, (2) many of these harms remain insufficiently supported by empirical evidence, and (3) certain categories of harm such as those related to social interactions and collaboration are still under-represented in the literature. This paper makes two primary contributions: (1) a taxonomy of generative AI harms specific to computing education, and (2) empirical insights into how these harms are currently framed and addressed in the field.

2 Methodology

The goal of this research is to understand the negative impacts and unintended consequences of GenAI in computing education. We conducted a systematic literature review (SLR) following the Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) reporting guidelines [170]. Below we discuss the details of this process, which are also summarized in Figure 1.

2.1 Scope of the Review

To ensure a focused and manageable review, we first define the scope of our search, including inclusion/exclusion criteria and temporal boundaries.

2.1.1 Inclusion and Exclusion Criteria. Over the last few years there has been a substantial increase in the number of GenAI papers [183, 184]. We adopted multiple exclusion criteria which are summarized in Table 1 to help us focus on the most relevant papers. We excluded papers that were not written in English, that did not focus on generative AI, and papers that were not specifically about computing education. To further reduce the scope, we focused on higher education and omitted k-12 research. We also excluded non-peer-reviewed materials, such as compiled conference proceedings and other informal publications.

2.1.2 Temporal scope and year reporting. Our searches covered 2022-2025 and concluded in July 2025. For any year-by-year plots, we intentionally omit 2025. Including 2025 would bias counts towards venues with early publication schedules (e.g., SIGCSE) and underrepresent later 2025 conferences whose proceedings publish after our cutoff (e.g., ICER, ITiCSE).

Table 1: Exclusion and inclusion criteria.

Exclusion Criteria
EC_1: Not available in English
EC_2: No clear generative AI focus
EC_3: Not related to computing education
EC_4: Focus is not about higher education (i.e: k-12)
EC_5: Not an academic paper (e.g.: conference proceedings)
EC_6: Does not contain evidence
Inclusion Criteria
IC_1: Discusses or demonstrates generative AI harms

2.2 Search and Selection Process

As mentioned, our process is guided by the PRISMA reporting guidelines, following three phases [170]. In the **Identification Phase**, we identified potentially qualifying citations by searching the ACM Digital Library, IEEE Xplore, and Scopus. During the **Screening Phase**, we filtered the resulting papers to ensure papers meeting the exclusion criteria were rejected. With the **Eligibility Phase**, we further filtered the set of references resulting from the screening phase to verify that the inclusion criteria was met.

2.2.1 Identification Phase. To ensure an overarching and comprehensive sample, we used three databases in our literature review: ACM Digital Library, IEEE Xplore, and Scopus. These databases were selected to align with those used in recent working group reviews on generative AI in computing education [183, 184]. ACM DL and IEEE Xplore include publications from prominent computing education conferences, such as SIGCSE Technical Symposium, ITiCSE, and FIE. These databases also include publications from ACM TOCE and other relevant journals. Scopus includes some overlap with both databases while also including some relevant journals that the other databases do not include.

- **`Domain:`** "computer science education" OR "computing education" OR "CS education" OR "CSEd" OR "CER" OR "programming education" OR "introductory programming" OR "CS1" OR "CS2" OR "computing student*" OR "CS student*" OR "computer science student*"
- **`Topic:`** "LLM*" OR "large language model*" OR "generative AI" OR "ChatGPT" OR "GPT" OR "Bard" OR "Gemini" OR "Claude" OR "GenAI" OR "Copilot" OR "artificial intelligence" OR "AI"
- **`Publication Date:`** Jan 2022 to Jul 2025

We combined these terms to find papers that focus on computing education and also involve large language models or generative AI.

`Search String` = **`Domain`** AND **`Topic`**

2.2.2 Screening Phase. As a first step in filtering the papers, we removed 151 duplicate papers and 21 papers that had been marked as 'retracted' by the conference or journal. After removing these 172 papers, we were left with 1,677 references.

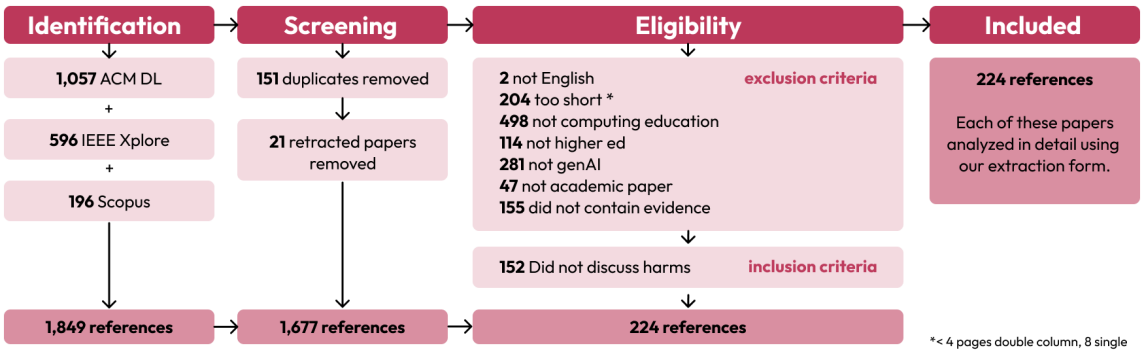

Figure 1: PRISMA diagram detailing our identification, screening, and assessment of paper eligibility for inclusion.

2.2.3 Eligibility Phase. In the second filtering pass, papers were excluded if they did not meet the inclusion criteria or violated any exclusion criteria (Table 1). Four researchers conducted this in two stages. In the first stage, the team applied the exclusion criteria, achieving an inter-rater reliability (IRR) of 0.84 using Fleiss' Kappa. In the second stage, papers were retained only if they met the inclusion criteria, with an IRR of 0.69. This process resulted in 224 papers retained and 1,453 papers removed.

2.3 Data Analysis

We conducted a deductive coding of each included study across several dimensions, drawing on established taxonomies where possible. The dimensions include contribution type, research methods, type of generative AI tool examined, and participant population.

2.3.1 Research Methods. We coded each study based on the reported research methods using a modified version of the taxonomy by Mack et al. [135]. Their taxonomy includes **controlled experiment**, **usability study**, **field study**, **interview**, **questionnaire**, **case study**, **focus group**, **workshop or design study**, **randomized control trials**. Based on our initial reading of the papers, we realized ethnographic methods and benchmarking were also common and we added those codes as well.

Studies could be coded as having multiple research methods to account for mixed-methods research. For example, there were multiple cases where interviews were used alongside surveys.

2.3.2 GenAI Tool Type. GenAI tools continue to evolve from the early language models to conversational agents, multimodal models, and specialized pedagogical tools that have been developed to explicitly support education. To capture this diversity, we coded for the following types of GenAI tools:

- **Chat Models** — Conversational agents such as ChatGPT, Claude, or Gemini.
- **Code Assistants** — Code-generation tools such as GitHub Copilot or Amazon CodeWhisperer.
- **Search-Integrated Models** — AI tools embedded within search engines, such as Perplexity or Gemini Search.
- **Image Models** — Generative image models such as DALL-E or Midjourney.
- **Custom or Specialized Tools** — Tools or prototypes that integrate generative AI for explicit educational goals.

As these tools continue to evolve and improve, it is helpful to know which harms have been associated with each type of tool. For example, some students face metacognitive challenges when using Code Assistants [187] and it is useful to understand whether and how these harms extend to other tools.

2.3.3 Task. Harms depend on the context of use and some might be more heavily associated with some tasks than others. We adopted the 'Context of Usage' taxonomy from Hou et al. [95], which describes four primary use cases for GenAI in computing education. These include **understanding course concepts**, **debugging code**, **identifying corner cases**, and **writing code**. This taxonomy was based on students perspectives, so we also included tasks from the instructors perspective including **assignment creation**, **image generation**, **data generation**, and **feedback**. These tasks were identified thematically as we read the papers during the filtering process (i.e.: screening and eligibility).

2.3.4 Population. A recent working group formed to explore how GenAI is used in upper-level courses [34], because, as they highlight in their literature review, most research about GenAI in computing education has focused only on the introductory courses. Therefore, we coded for the following student populations:

- **Introductory undergraduates** — students enrolled in CS1 or other entry-level courses
- **Upper-level undergraduates** — students enrolled in undergraduate courses beyond the introductory sequence
- **Graduate students**
- **Instructional staff** — including instructors, teaching assistants, or course designers
- **Unspecified undergraduates** — studies that involved undergraduate students but did not specify course level

2.3.5 Levels of Evidence. During our initial reading, we observed that many papers described harms hypothetically and few papers provided explicit evidence for the harms that they described. For example, instructors might **hypothesize** that their students would develop misconceptions when viewing incorrect code generated by an AI coding assistant. While this is a plausible concern, it is presented speculatively, without empirical evidence. In cases where evidence was provided, it ranged from informal **observations** to more rigorous forms of **measured** evidence. We adopted a broad

Figure 2: Stacked bar chart showing the percentage of papers that discussed harms of GenAI in computing education. Greyed bars represent papers not analyzed. *Papers from 2025 and other excluded works are omitted.*

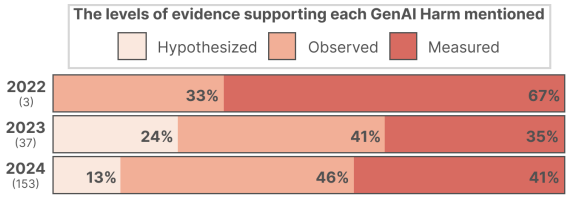

Figure 3: The stacked bar chart shows the proportion of papers based on their highest level of evidence. *Papers from 2025 are omitted from this visualization*

definition of measured evidence, including not only controlled experiments but also **surveys, interviews, and benchmarking studies.** It is important to note that we explicitly coded each paper for the highest level of evidence it provided for any harm instance in the paper, and only one label was applied for each paper.

2.3.6 Types of Harms. To identify the types of harms, we first familiarized ourselves with the data by open-coding [232] a randomly selected subset of the papers. As we continued coding the data throughout the filtering process, we expanded the codebook in the rare cases where a new harm was identified. When new codes emerged, we retroactively reviewed previously coded papers to determine whether these newly identified harms were present. We began coding with a very fine-grained coding scheme to capture very specific harms. In later stages, we combined codes and established themes to fit the data. This inductive process enabled us to identify unexpected harms and we exercised reflexivity [35] throughout the process to reflect on how our positionality (see Section 3) might impact our analysis. These harms and sub-harms are described along with the coded papers in Table 2 and Table 3.

3 Positionality Statement

Members of our research team have been involved in studies that have investigated both the benefits and challenges of AI tools in educational contexts. Our prior research focused primarily on the benefits of GenAI, but also identified a few concerns. Consequently, we did not approach this review with the belief that AI tools are inherently harmful or that their use in education should be avoided. Rather, we recognize generative AI as a transformative technology that is rapidly becoming a central aspect of society, industry, and education in the future. Given this transformative impact, we also recognize the importance of understanding how this technology impacts students and how we might shape the technology and our pedagogies to better support our students.

Our goal for conducting this review is not to discourage the use of AI tools but to identify risks and harms so they can be understood, addressed, and mitigated. We believe that a thorough assessment of these risks is essential to maximize the benefits.

In addition to these perspectives, our team also contains a mix of junior and senior researchers, with some members who are undergraduate students. This perspective as educators, researchers, and students further informed our analysis.

4 Results

As shown in Figure 2, there has been a consistent increase in the number of papers published about GenAI in computing education with 2024 accounting for more papers than the previous two years combined. The number of papers mentioning harms has also consistently increased from 2022, and this increase is disproportionately higher relative to the overall rise in papers discussing GenAI in computing education. This suggests that as GenAI becomes more prevalent in computing education, researchers are also becoming more aware of potential harms and pitfalls as well.

However, Figure 3 shows a less pronounced trend in the reporting of evidence, with only a modest increase observed between 2023 and 2024. Caution should be exercised when interpreting the 2022 data, as the sample size for that year is too small to support interpretations of trends.

4.1 The types and prevalence of harms

As shown in Figure 4, we identified six broad categories of harm, with multiple sub-harms. Some harms, such as those about cognition and metacognition appeared more frequently in the literature than others, such as social harms involving peer interactions and social dynamics. There were also differences in the types of evidence used to demonstrate harms. For instance, concerns about cheating and plagiarism were often supported through speculation and argumentation, rather than empirical evidence.

For less frequently mentioned harms or those with limited empirical support, this may reflect that they are indeed less prevalent or considered less important within the current research landscape. However, it may instead reflect methodological challenges for studying these harms. In the following subsections, we provide examples of the harms we identified in the literature.

4.1.1 Cognitive. The most prevalent umbrella harm (32.4%) across the papers was related to Cognition. These harms typically concerned how students acquire new knowledge or apply existing knowledge to problems. This category also had the highest proportion of evidence-based harms (28.2%), suggesting that concerns about cognitive impacts are not only common but also more frequently supported by empirical data.

Acquiring Knowledge. Of the studies that discussed Cognitive harms, most of them (86.2%) focused on ways that GenAI harmed students' ability to acquire knowledge. Many papers reported that GenAI produces inaccurate responses, or hallucinations [24, 32, 94, 153, 218], with one study noting that a third of

Table 2: Cognitive and Metacognitive Harms identified in the SLR

Harm	Definition	Relevant Papers
Cognitive		
Acquiring Knowledge	Incorrect information (i.e., Hallucinations) or shallow learning (e.g.: copy and pasting code)	[1, 3–8, 10, 12–16, 18–28, 31–33, 36–39, 39–43, 45, 47–50, 52, 58, 59, 59–61, 64, 65, 67, 70, 70–74, 76, 76–79, 82–92, 94, 95, 97–100, 100–102, 104–108, 110, 111, 113–119, 121, 123–131, 131–134, 136, 139, 140, 142, 144, 146, 148, 150, 151, 153–155, 155, 157, 157–160, 162–168, 171, 173–176, 178–180, 182–185, 187, 189, 190, 193, 193–199, 201, 203, 205–207, 209–215, 218–227, 227, 229, 233, 235–238, 241, 242, 244, 244–251, 251–258, 258]
Applying Knowledge	Decline in problem-solving, coding, writing skills, homogenization of code styles.	[5, 14, 17, 22, 23, 33, 37, 39, 41, 45, 47, 52, 53, 58–61, 64, 73–75, 82, 86, 92, 110, 111, 114, 119, 134, 144, 146, 162, 167, 173, 175, 187, 193, 210, 219, 220, 222, 224, 226, 231, 233, 235, 236, 244, 250, 253, 254, 258]
Metacognitive		
Disruptions to Metacognition	Negative impacts on students' ability to develop a plan for solving a problem, monitor the plan, and adapt the plan as necessary.	[2, 5, 6, 8, 13–18, 20–23, 26, 28, 31–33, 36, 39–41, 45–47, 50, 53, 58, 61, 65, 67, 72–74, 77, 79, 81, 82, 84–89, 89–91, 94–97, 99, 101, 102, 104–108, 110, 111, 113–119, 121, 123, 124, 126, 129–131, 131–134, 136, 138, 140, 146, 148, 150, 153, 159, 162, 164, 167, 168, 168, 178–180, 183–185, 187, 189, 190, 193, 193, 206, 209, 212–215, 219–224, 226, 227, 229, 230, 233, 235, 236, 240–242, 246, 248–251, 253–256, 258]
Over-reliance and trust calibration	Students either over-rely on GenAI or misjudge how much to trust its outputs.	[1, 2, 10, 12, 14–17, 19, 21–23, 31, 37, 39–41, 48–50, 62, 64, 66, 70, 72, 75–77, 82–84, 86, 92, 95, 98, 106, 114, 116–118, 127, 131, 131, 133, 134, 136, 139, 142, 153, 153, 162, 168, 171, 184, 187, 193, 193, 206, 209, 211, 211, 213, 219, 220, 222, 223, 227, 233, 245, 249, 251, 252, 254, 257]

Table 3: Classroom Harms identified in the SLR

Harm	Definition	Relevant Papers
Assessment and Academic Integrity		
Cheating & Plagiarism	Includes AI-enabled cheating and academic dishonesty.	[4, 6, 16, 17, 20, 21, 23, 26, 28, 31, 32, 37, 38, 41–43, 45–47, 49, 53, 57, 58, 60–63, 74, 77, 83–88, 90, 91, 94, 96–99, 102, 106–108, 110, 111, 114, 118, 119, 123, 130, 132, 132, 140, 141, 144, 146, 148, 150, 152, 164, 168, 171, 173, 183–185, 188, 190–192, 202, 210, 212, 214, 215, 219, 222–225, 227, 228, 231, 233, 236, 241, 244–246, 248, 250, 254–256, 258]
Fairness & Enforcement	Ambiguous policies, unfair accusations without proof, and inconsistent enforcement across instructors and institutions	[6, 6, 10, 11, 11–13, 22, 23, 38, 43, 45, 46, 50, 57, 59–61, 63, 71, 74, 83, 83, 85, 85, 87, 88, 91, 94, 94, 97, 97, 98, 102, 110, 110, 110, 119, 119, 131, 132, 140, 140, 140, 141, 141, 141, 141, 144, 145, 145, 146, 148, 149, 149–152, 161, 167, 184, 185, 188, 188, 192, 202, 202, 210, 210, 211, 214, 215, 215, 218, 219, 235, 240, 242, 246–248, 248, 252, 255]
Equity		
Model biases	Biases in GenAI responses related to language, race, and gender.	[3, 16, 25, 32, 41, 62, 66, 84, 90, 98, 99, 105, 106, 108, 114, 118, 125–127, 129, 130, 143, 154, 164, 164, 167, 172, 172, 183–185, 185, 186, 207, 226, 227, 230]
Access & Opportunity	Unequal access to AI tools or uneven distribution of benefits.	[19, 41, 51, 62, 62, 63, 65–67, 76, 77, 82, 84, 88–90, 94, 95, 97, 99, 103, 105, 105–108, 114, 117–119, 126–130, 132, 153, 153, 154, 154–156, 164, 171, 172, 183–186, 188, 195, 198, 203, 205–207, 212, 219, 220, 251]
Social Harms	Reliance on GenAI can reduce collaboration and increase feelings of isolation.	[19, 31, 40, 41, 90, 95, 97, 102, 119, 146, 152, 169, 171, 171, 172, 177, 184, 216, 226, 228, 241]
Instructional and Logistical	Logistical challenges, privacy concerns, and new learning objectives	[8, 15, 17, 19–21, 36, 37, 45, 46, 51, 53, 57, 59, 66, 76, 82, 84, 86, 90, 94, 95, 97, 99, 103–108, 114, 117–119, 121, 123, 124, 126, 126–128, 131, 131–133, 138, 139, 142, 144, 145, 147, 148, 150, 154, 156, 159, 168, 171, 182, 184, 187, 193, 193, 203, 206, 209, 221, 222, 233, 236, 241, 242, 245, 246, 248, 251–253, 255, 256]

student-AI interactions led to wrong or incomplete answers [17]. Nguyen et al. shows that as students begin to adopt GenAI as a teaching tool, its feedback can be uneven, sometimes detailed and supportive, but also occasionally inaccurate or less effective for deep learning compared with instructor feedback [161]. Students also describe the risk of shallow learning, stating that AI can make work feel 'too easy', raising concerns about overdependence and reduced effort [37].

In a data structures and algorithms setting, Palacios-Alonso et al. found that when TAs guided students' prompts and verified ChatGPT's outputs, the AI-assisted group outperformed the control by 16.50 points on average [171]. However, for more complex algorithms (e.g, Dijkstra's, tree diagrams) outputs were sometimes inaccurate so TA checking was still required [102].

Poitras et al. found that lower-proficiency learners (reading 71.8% vs. 78.7%, writing 77.7% vs. 83.9%, both $p < .05$) relied on GenAI in 67% of submissions primarily for generation (51%), revision/debugging (44%), and explanations (15%). However, they did not outperform manual coders (no significant assignment-score effects; $R^2 = 0.03$) [178]. Similarly, in a controlled student experiment with Code Defenders, a gamified web platform for practicing software testing via mutants, students who relied more on the assistant produced 8.6% fewer tests, 78.0% of which were not useful [148].

Applying Knowledge. Papers that related to applying knowledge tended to focus on students getting code to run with less independent problem solving, alongside shifts in code style, reduced exploration, and weaker engagement in critical thinking. A CS2 study reported a statistically significant rise in atypical code styles after widespread adoption of ChatGPT, from 12% of submissions in 2022 to 30% in 2023 [50]. Students also described feeling 'lazy' to explore alternatives, saying that GenAI tools limited their inclination to try new ideas [226]. In a semester-long field study of CodeTutor, an LLM-powered course assistant, students reported that the tool was useful for syntax help and debugging. Yet many noted it did little to foster deeper critical thinking, often encouraging reliance on quick solutions over problem analysis [132].

A recent study also showed that in about a third of cases, students attempted to complete assignments by pasting the full task description into ChatGPT and proceeding with little or no independent work [17]. Teams sometimes adapted by decomposing complex prompts into smaller subproblems to obtain more usable outputs to address the challenges where GenAI produced unhelpful responses for harder tasks [20].

4.1.2 Metacognitive. The second most frequent category of harms was negative impacts on students' ability to develop a strategic plan, monitor the plan, and adapt it as needed.

Disruption of metacognition. Many of the studies reviewed described the potential for disrupting students' ability to plan and to monitor their problem-solving process with quite a bit of evidence to support claims of over-reliance or over-trust in AI assistance. A paper by Scholl et al. reported that students couldn't tell if they understood responses from AI assistants or whether they were simply parroting the models [221]. In their CS1 deployment, they analyzed 213 valid student chat logs and found an average of 10.96 prompts (median 7). Shorter sessions (0–5 prompts) were dominated by direct solution requests (81% of prompts), while longer sessions (12+ prompts) showed more mixed behaviors (41% direct requests). This pattern reflects a shift toward "prompt-and-accept" workflows—asking for and accepting solutions—rather than engaging in planned, monitored problem solving [221].

Related work by MacNeil et al. similarly found that students were more likely to seek direct solutions as deadlines approached [55], and Kapoor et al. reported that 50% of students disabled guardrails to reveal full solutions, often citing time pressure or lack of self-regulation as motivations [109].

Prather et al. also observed metacognitive challenges through an eye tracking study of how students use Github Copilot. Students with lower grades exhibited more AI-specfic difficulties and took longer to complete the task with new difficulties negatively correlated with course grade ($r = -0.503$, $p = 0.020$) and positively with time to completion ($r = 0.693$, $p = 0.0005$). These students accepted a larger share of Copilot suggestions (34.1% on average) than peers without metacognitive issues (24.5%), and were more often misled or distracted by the suggestions [187].

Rogers et al. captured the mixed attitudes that this can create. When asked whether "ChatGPT could hinder my ability to learn how to program for myself", responses were bimodal, with disagreement most common (39.7%) but agreement the second most common (25.0%; strongly agree 8.8%, undecided 13.2%, strongly disagree 13.2%) [208].

Some papers also demonstrated that students were also concerned themselves about over-reliance, saying that:

> *"It's helping as well, but at the same time, it's reducing the human power, and human thinking ability. Like we always depend on ChatGPT for new ideas. We are not running our own minds."* [37]

Over-reliance and trust calibration. Many papers describe students either over-trusting model output or leaning too heavily on it. For example, a paper by Alves and Cipriano [15] found that 47.2% of students accepted one of the solutions provided by ChatGPT, and 72.2% incorporated a solution fully or partially into their project code. Students also try to reduce the individual effort they put in, in a CS1 case study, about a third of attempts consisted of submitting the full task to ChatGPT with no changes [17]. Additionally, in an HCI written exam where ChatGPT use was permitted, some students reported defaulting to copying answers completely [70].

This polarization also appears in trust calibration. One survey reports 16% distrust, 36% neutral, and 47% trust (N=253), indicating sizable groups at risk of either over- or under-reliance when planning and monitoring their work [17].

Students themselves articulate this risk. In reflections from an ethics assignment, one group warned that novices may not detect their AI mistakes and urged validating outputs against other sources [77]. Oyelere et al. reported a positive correlation between using AI for complete solutions and the belief that knowing how to program is unnecessary ($r = 0.321$, $p < 0.001$) [167]. Faculty also share these concerns, Azoulay et al. argue that the ease of AI solutions can distance independent problem solving [23].

In reflections around a custom tool, CodeHelp, some students admitted to going straight to the tool and feeling too dependent on it [126]. Connections can again be drawn to Scholl et al. and their work analyzing ChatGPT logs of a CS1 course, finding students often parroted outputs from genAI rather than building upon them and understanding [220].

4.1.3 Assessment and Academic Integrity. As shown in Figure 4, harms related to Assessment and Academic Integrity were the third most commonly identified in our review. The most common subharm within this category was related to concerns about cheating and plagiarism, with a few papers also considering harms related to fairness and enforcement.

Cheating and plagiarism. Most papers about cheating and plagiarism did not present measured evidence (8.4%) for the harms they described. Some of the harms focused on the factors that led students to engage in academic misconduct using GenAI. For example, students tended to rely on AI when facing time pressures [21, 111] or when they are not motivated by the work [21, 31, 114, 254]. These findings largely replicate what is previously known in research on plagiarism and academic dishonesty [9].

Researchers also developed benchmarks to determine how effectively AI models could be used to solve various computer science assignments [45–47, 56, 80, 94, 130, 199, 217, 218] with a framing around academic misconduct. Early work with Codex showed promising but imperfect performance: on two CS1 exams, Codex scored about 78% on each and averaged roughly 50% across Rainfall

variants, struggling with constraints and formatting [56, 68]. However, subsequent papers quickly showed large gains with newer models. For example, an ITiCSE working-group replication and follow-up analysis reported GPT-4 achieving 99.5% and 94.4% on two CS1 exams and solving all Rainfall variants [56, 183]. Recently, multimodal models can solve assignments, such as data structures and algorithms problems, based only on an image [81, 94].

Across these papers, it is clear that GenAI can now solve most programming problems that students encounter at various stages of their education. This quote from an interview study of faculty experiences by Sheard et al. [223] summarizes it well: *"I have lost my ability to confidently assess any work that students hand in."*

Fairness and Enforcement. Given these concerns about academic integrity, some of the papers discussed new considerations for assessment, including how to prevent or detect cheating, and a few discussed issues of fairness. In terms of enforcement, most papers focused on detection or prevention. The detection papers included identifying stylistic markers of AI use [57] or the use of advanced concepts which had not been taught [159]. Additional detection approaches have included code-anomaly detectors that flag unusual syntax patterns compared to student's prior work, which can help identify AI-assisted submissions. Their accuracy, however, drops in collaborative contexts [91]. While promising, these methods raise fairness concerns if used as sole evidence in high-stakes enforcement.

Further studies propose preventative approaches to curb academic dishonesty, such as Prompt Problems, which present problems visually to prevent simple copy-pasting of design briefs [53]. Additional work evaluates the qualities of assignments that makes them difficult to be solved directly by GenAI [144].

Other work suggested that instead of preventing cheating, some faculty are embracing AI use [23, 119, 152, 254], acknowledging that 'resistance is futile' and that students who want to cheat will continue to be able to do so [119].

Yet even when AI is allowed in courses, there are still unequal effects that emerge. Many students paste AI answers during assignments or exams, sometimes gaining an advantage but often being misled by incorrect responses, which raises equity concerns about who benefits and who is disadvantaged [110].

Fairness issues also arose when AI is used on the instructional side. In blinded studies, students struggled to distinguish ChatGPT-generated feedback from that of human TAs, sometimes even accepting weaker AI feedback at face value [43]. Additional work highlights the wide spectrum of faculty enforcement stances, from outright bans to pragmatic acceptance of AI use, which raises concerns about inconsistent policies across courses and institutions [120].

4.1.4 Equity. Concerns about equity appeared in 29% of papers in our review. These concerns largely focused on issues with the models or inequitable access to these models for computing students.

Model biases. Several studies described harms associated with various biases, including reports of poorer performance for low-resourced languages and the reinforcement of harmful stereotypes and prejudices. A few papers focused on multilingual support. Prather et al. showed mixed findings with some students preferring to write code in their own native language, and others claiming that GenAI has better support for the English language [186]. Jordan et al. similarly demonstrated that GenAI is more accurate in when prompted in some languages than in others [105].

Biases were also present in text-to-image models. Apiola et al. taught a module to have CS students explore social justice issues in text-to-image models and found that "the most common type of harm identified by students was the harm of application of stereotypes or prejudices" [19]. These representational biases have the potential to reify existing biases and stereotypes that are common within computer science [44, 234].

Access & Opportunity. A few papers also identified harms related to access and opportunity. These harms included uneven usage of GenAI tools, uneven benefits, and practical barriers to participation. Early studies that investigated GenAI usage showed a bimodal distribution between students who used GenAI frequently and those who never used GenAI [95, 114, 183].

Beyond usage gaps, barriers include language support, subscription costs, and hardware requirements. For multilingual contexts, model accuracy and problem quality vary by language, advantaging English over some lower-resourced languages [105, 186]. Cost and infrastructure can also exclude students or institutions [127]: for example, running OpenAI models at scale incurs nontrivial recurring costs, and local hosting requires capable GPUs. A rough estimate is $5–$10 per 1,000 short conversations with GPT-3.5 versus $250–$500 with GPT-4, and a server-side GPU such as an RTX 3090 (24 GB VRAM) for multi-model deployments [127]. Beyond these institutional costs, students themselves have reported barriers such as subscription fees and device limitations that limit their ability to experiment with GenAI tools compared to peers who can afford paid access [82]. Faculty have also noted that licensing costs and uneven institutional resources mean some programs are unable to integrate these tools into coursework [171], and others highlight that institutions without budgets for subscriptions or hardware upgrades face significant disadvantages [172].

In addition to these differences in usage, Lyu et al. found that students with prior GenAI experience tended to benefit more than those without [132]. Similarly, Prather et al. demonstrated that students who were doing well in the course did even better when using Github Copilot where students who were struggling did worse [187].

4.1.5 Social Harms. Only 19 papers (8.5%) highlighted harms of generative AI on social aspects of learning, the least frequently referenced harm in our literature review. These harms included students feeling pressured to use ChatGPT in order to keep up with their peers [169], feeling their social interactions being affected by GenAI [95], and feeling guilty about their AI usage [171].

Hou et al. first demonstrated the potential for social harms with examples of how students relied on ChatGPT to avoid the socio-emotional barriers of help-seeking [95]. They included quotes from participants such as the following:

> *"It is easier for me to ask ChatGPT because I don't have to worry... I feel like it is kind of rude to continuously ask for more and more details."*

The following year, Arora et al. shared that three participants *"noted that LLMs actually hindered collaboration due to compatibility issues between LLM-generated code from different team members"* [20],

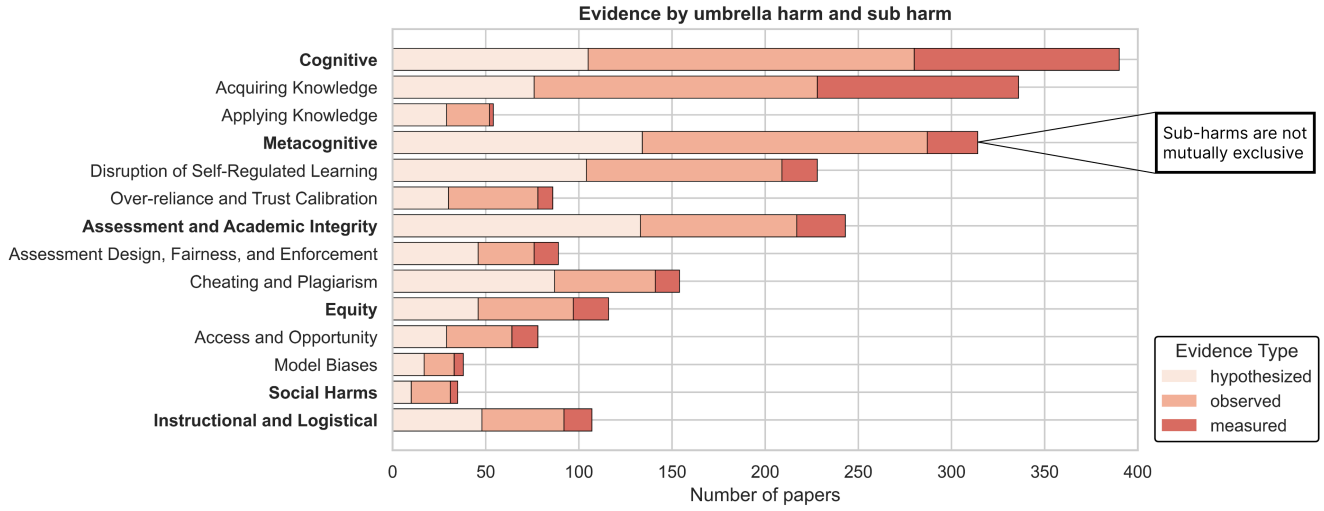

Figure 4: Umbrella harms (e.g., Metacognitive) are bolded and represent the total number of instances across all associated sub-harms. Because sub-harms are not mutually exclusive, a single paper may be counted multiple times within an umbrella category if it includes multiple sub-harms with the same type of evidence.

highlighting how mismatched outputs can complicate integration and slow teamwork.

4.1.6 Instructional and Logistical. Some of the studies identified ways that GenAI created logistical challenges for instructors or required them to make substantial changes to their teaching practices. For example, in one study, students raised concerns about how to protect their data when using chatbots [193]. Another study reported similar privacy concerns, noting that one student refused to create a ChatGPT account due to fears about data use [248]. In some countries, student data is protected (e.g.: FERPA in the USA) or broader protections exist for citizens more generally (e.g.: GDPR in Europe). These privacy concerns were seldom discussed.

In some studies, concerns were also raised about the negative impacts AI has had on students' motivation. For example, students described moments where learning felt 'too easy' or like 'cheating,' which undermined motivation to wrestle with problems and reduced the value they placed on instructor feedback [37]. At the same time, several papers pointed to an instructional gap that students can run certain tools, but often lack the skills to craft effective prompts or evaluate answers. In a semester-long study, 63% of student prompts were rated unsatisfactory, reinforcing the need to build GenAI literacy intro courses [132]. Budhiraja et al. also noted reduced motivation alongside a need for explicit guidance on prompt engineering and effective use [37]. Many expect AI to become integral to professional software practice, arguing that curricula should teach responsible and proficient use, even as some worried about job displacement [120, 208]. Instructors also emphasized preparing students for AI-mediated workplaces by integrating these tools into coursework [120].

Translating these needs into practice requires structure. Recent work cautions that switching to visual or proctored assessments alone does not solve the issue. AI proctoring, in particular raises accessibility, privacy, and autonomy concerns. As a result, instructors are rethinking both assessment design and how students are guided when using GenAI [94]. LLM-based recommendations were sometimes repetitive, out of context, or futile, which students found demotivating [159]. Consequently, instructors must take a more active role in scaffolding the process such as through the use of guardrails that encourage students to use tools responsibly [53, 112, 126, 137].

Finally, multiple studies described how AI is changing the relevance of various skills and competencies [245]. For example, instructors now may need to focus more on problem specification, code comprehension, and problem decomposition [144]. Some new skills are also required such as prompt engineering [53].

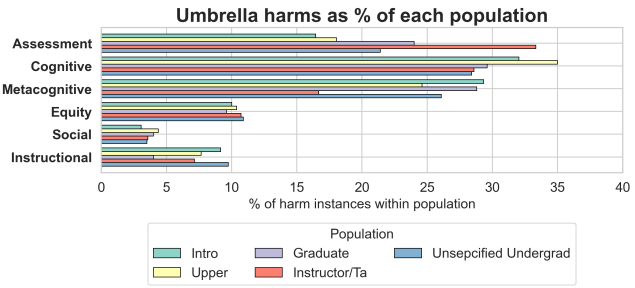

Figure 5: Harms as a percentage of each population. 69.6% of instances involved introductory students, 12.0% upper-level undergraduates, 9.3% graduate students, and 9.1% educators

4.2 The Contexts of Harms

4.2.1 Population. To better understand who is most impacted by harms, we categorized harm instances by the targeted learner population. Most papers focused on introductory students, followed by unspecified undergraduate groups (where the population was

only described as undergraduates), and upper-level students. Harms relating to cognitive and metacognitive challenges were most prevalent for these groups. For example, intro-level students accounted for 189 cognitive harms and 173 metacognitive harms, which was much larger than other populations.

Assessment-related harms were more evenly distributed with notable instances being reported among graduate students (n=30) and instructors/TA's (n=28). Equity harms were also cited across populations, however, they most prominently appeared among intro and unspecified undergraduate students.

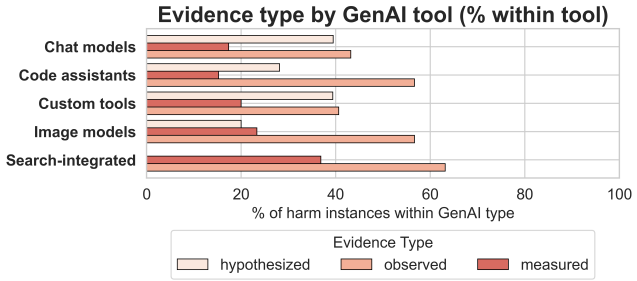

Figure 6: Percentage of harm instances by evidence type within each type of GenAI tool.

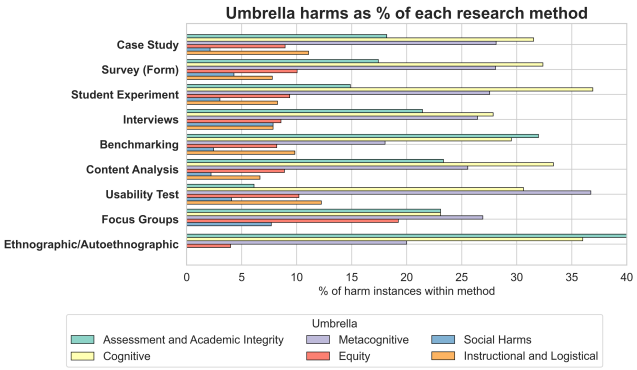

Figure 7: Umbrella harms across research methods.

4.2.2 GenAI Tool Type. We also examined how different GenAI tools mapped onto different harm categories Figure 6. Chat models (eg. ChatGPT, Claude, Gemini) were the most common across all harm types, accounting for 214 instances of harms related to academic integrity, 323 cognitive, and 266 metacognitive. This most likely shows the widespread and general purpose use of chat models making them relevant for a range of student activities.

Code assistants (e.g, Github Copilot, CodeWhisperer) were the second most commonly discussed tool type. The most frequently reported harms associated with these tools were cognitive (n=79), academic integrity (n=53) and metacognitive (n=59). This aligns with their primary use case being code generation, which could impact how students think through programming tasks.

Custom or specialized tools (e.g., course-specific prototypes or research-built interfaces) were most commonly linked to cognitive (n=62) and metacognitive (n=46) harms, which could be a result of targeted experiments with these custom tools.

Image models (eg. Dall-E, Midjourney) and search-integrated tools (e.g., Perplexity, Gemini Search) appeared far less frequently. These tool types were occasionally connected to equity or social harms, but their overall harm counts were much lower, likely a result of their more limited scope in computing coursework.

4.2.3 Research Methods. As shown in Figure 7, harms were not uniformly distributed across research methods. Methodological choices may have shaped the types of harms that researchers identified, as some methods may surface certain harms more readily than others. For example, cognitive harms were the most frequently reported category, appearing in 36.9% of student experiments, 33.3% of content analyses, and 27.9% of interviews. Metacognitive harms were also common, most visible in surveys (28.1%) and interviews (26.4%). Benchmarks primarily highlighted assessment and academic-integrity harms (36.8%).

5 Discussion

Our systematic literature review provides a useful and timely overview of the harms that have been hypothesized, observed, or measured by researchers and practitioners. Our results can be contrasted with a recent working group by Prather et al. where they reviewed the literature on specific GenAI interventions and systems [184]. In that work, they reviewed 71 papers and focused on studies that featured an explicit intervention. They found 5 studies which shared negative outcomes compared to 46 positive outcomes, 15 mixed outcomes, and 5 instances which were neutral. This review provides a much more comprehensive view, focusing on 224 papers and not just papers with instructor interventions. We found 16.3% of the papers discussed harms compared with 7.0% in that prior work. This may reflect a positivity bias for researchers studying their own interventions or it may be explained by a growing emphasis on negative papers. In either case, our findings show the breadth and depth of harms within computing education.

We also observed a bias toward certain types of harms when compared with others. Cognitive and metacognitive harms were rather prevalent, but other harms such as negative social impacts of GenAI were much less prevalent. Although a few anecdotal accounts suggested that students were relying on AI instead of their peers and interfering with collaboration. A recent paper by Hou et al. provides a much more comprehensive view into the ways that AI is disrupting peer interactions and negatively impacting learning communities [93]. Another paper discusses students preferring to ask ChatGPT questions instead of asking their professors or TAs, in fear of judgment or asking a dumb question, despite acknowledging ChatGPT's inaccuracy [20]. For other harms, such as equity, which also received limited attention, there was very limited discussion about the impacts on eventual careers, society, or the environment. This shows how the harms are still being understood and suggests that more research is needed to understand the harms beyond the well-studied harms related to assessment, cognition, and metacognition.

5.1 Grand Challenges

Our taxonomy of harms, which includes assessment, cognition, metacognition, equity, social, and logistical categories, serves as an important road map or grand challenges to guide future research. As future work, researchers should consider the following questions:

5.1.1 What is the future of assessment? There appears to be growing evidence that GenAI can solve most programming problems which makes it hard to assess student outputs directly. One proposed solution has been to assess process instead of product. However, as GenAI tools increasingly support process-level tasks (e.g.: problem decomposition), this too may remain a challenge.

5.1.2 What cognitive harms remain as AI models improve? While many of the cognitive harms in the papers were attributed to hallucinations or errors, these are being rapidly addressed by model improvements. However, how do we ensure students continue to build foundational problem-solving skills when AI appears to 'solve' the problems for them?

5.1.3 How can we mitigate metacognitive harms? Perhaps one of the most concerning harms is the "illusion of competence" that GenAI models offer to students [187]. Students are misled and don't realize it, but worse, they feel productive. One recent approach helps students identify misalignments between their intentions and the AI's plan by simply offering multiple options for each code suggestion [137]. How do we help students calibrate their trust and identify these misalignments?

5.1.4 How can we ensure everyone is learning effectively? Multiple papers showed uneven usage and unequal benefits for different student groups. As researchers continue to understand cognitive and metacognitive harms, it becomes clearer that these harms are not experienced uniformly. Despite early promises of democratizing education, GenAI tends to benefit students who are already well-prepared, while leaving others behind.

5.1.5 How can we preserve our learning communities? While few papers addressed social harms, recent work has raised more serious concerns [93]. Learning is inherently social and sense of belonging has been consistently linked to retention [239]. How do we preserve peer interactions, collaboration, and friendships within the class?

5.1.6 What new logistical challenges must we face? GenAI has introduced a litany of new challenges for educators, including identifying new competencies, addressing data privacy concerns, scaffolding students' use of AI tools, and adapting to changing student motivation and preparedness.

These questions represent the grand challenges currently facing educators and are the direct result of harms students and instructors experience due to the introduction of GenAI. While there are many exciting possibilities going forward, our community should not lose sight of these challenges as well.

6 Limitations

Our review captures the current state of research on generative AI harms in computing education. However, this review only captures the harms that have been studied, and it is possible that students experience these harms to greater extent than the review might suggest. For example, the studies we analyzed examine harms associated with particular types of tools, but it is possible that some tools have received more attention than others. Some harms may appear more prominently in the literature not because they are more widespread or severe, but because they have been the subject of focused study. Conversely, the absence of evidence for certain harms in relation to specific tools should not be taken as evidence that such harms do not occur.

We also caution readers against interpreting our findings as suggesting that GenAI tools only harm students. While this review focuses on potential risks and negative impacts documented in the literature, GenAI also offers many opportunities to enhance student learning and engagement. Recognizing harms is a necessary step toward designing fair and effective educational experiences, but they are not a reason to reject or avoid AI altogether.

Finally, this is a scoping literature review with the goal to identify the primary types of harms and their relative prevalence. While we report on the types of evidence presented in the literature, a more rigorous investigation, such as a meta-analysis, could be a valuable future contribution to understand not just the types of harms but the magnitude of their impacts.

7 Conclusion

In this paper, we present the results of a systematic literature review that scopes the harms of GenAI in computing education. By carefully analyzing 224 that were chosen based on our inclusion and exclusion criteria, we found an encouraging trend that researchers are not only focusing on the benefits but also on the harms. However, this number is still quite low with only 21% of papers discussing harms in 2024. Our findings also show that some harms such as those related to impacts on cognition, metacognition, and assessment are more common, other harms were not as prevalent. Our work shows that despite many benefits, there is also a real potential for harm when educators do not consider the broader tool limitations or classroom contexts.

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References

[1] Willem Aerts, George Fletcher, and Daphne Miedema. 2024. A feasibility study on automated sql exercise generation with chatgpt-3.5. In *Proceedings of the 3rd International Workshop on Data Systems Education: Bridging education practice with education research*. 13–19. https://doi.org/10.1145/3663649.3664368

[2] Umair Z Ahmed, Shubham Sahai, Ben Leong, and Amey Karkare. 2025. Feasibility study of augmenting teaching assistants with ai for cs1 programming feedback. In *Proceedings of the 56th ACM Technical Symposium on Computer Science Education V. 1*. 11–17. https://doi.org/10.1145/3641554.3701972

[3] Zishan Ahmed, Shakib Sadat Shanto, and Akinul Islam Jony. 2024. Potentiality of generative AI tools in higher education: Evaluating ChatGPT's viability as a teaching assistant for introductory programming courses. *STEM Education* 4, 3 (2024), 165–182. https://doi.org/10.3934/steme.2024011

[4] Yutong Ai, Maya Baveja, Akanksha Girdhar, Melina O'Dell, and Andrew Deorio. 2024. A Custom Generative AI Chatbot as a Course Resource. In *2024 ASEE Annual Conference & Exposition*.

[5] Muhammad Fawad Akbar Khan, Max Ramsdell, Ha Nguyen, and Hamid Karimi. 2024. Human Evaluation of GPT for Scalable Python Programming Exercise Generation. In *2024 IEEE 11th International Conference on Data Science and*

Advanced Analytics (DSAA). 1–10. https://doi.org/10.1109/DSAA61799.2024.10722841

[6] Gokhan Akcapinar and Elif Sidan. 2024. AI chatbots in programming education: guiding success or encouraging plagiarism. *Discover Artificial Intelligence* 4 (2024), 87. https://doi.org/10.1007/s44163-024-00203-7

[7] Erfan Al-Hossami, Razvan Bunescu, Justin Smith, and Ryan Teehan. 2024. Can language models employ the socratic method? experiments with code debugging. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 1*. 53–59. https://doi.org/10.1145/3626252.3630799

[8] Rawan Al-Nsour. 2024. AI Tools in Matlab Course Education: Instructor Point of View. *Journal of Computing Sciences in Colleges* 40, 2 (2024), 95–104.

[9] Ibrahim Albluwi. 2019. Plagiarism in programming assessments: a systematic review. *ACM Transactions on Computing Education (TOCE)* 20, 1 (2019), 1–28. https://doi.org/10.1145/3371156

[10] Rayed AlGhamdi. 2024. Exploring the impact of ChatGPT-generated feedback on technical writing skills of computing students: A blinded study. *Education and Information Technologies* 29, 14 (2024), 18901–18926. https://doi.org/10.1007/s10639-024-12594-2

[11] Murtaza Ali, Prerna Rao, Yifan Mai, and Benjamin Xie. 2024. Using benchmarking infrastructure to evaluate LLM performance on CS concept inventories: Challenges, opportunities, and critiques. In *Proceedings of the 2024 ACM Conference on International Computing Education Research-Volume 1*. 452–468.

[12] Murtaza Ali and Benjamin Xie. 2025. Automated Benchmarking Infrastructure: Moving Toward Robust Investigations of Gen AI in Computing Education. *ACM Inroads* 16, 2 (June 2025), 46–51. https://doi.org/10.1145/3734876

[13] Umar Alkafaween, Ibrahim Albluwi, and Paul Denny. 2024. Automating Autograding: Large Language Models as Test Suite Generators for Introductory Programming. *Journal of Computer Assisted Learning* 41, 1 (2024), e13100. https://doi.org/10.1111/jcal.13100

[14] Eman Abdullah AlOmar. 2025. Nurturing Code Quality: Leveraging Static Analysis and Large Language Models for Software Quality in Education. *ACM Transactions on Computing Education* (2025). https://doi.org/10.1145/3722229

[15] Pedro Alves and Bruno Cipriano. 2025. "Give Me the Code": Log Analysis of First-Year CS Students' Interactions with GPT. In *Proceedings of the 17th International Conference on Computer Supported Education* (Porto, Portugal). SCITEPRESS - Science and Technology Publications, 198–207. https://doi.org/10.5220/0013194400003932

[16] Matin Amoozadeh, David Daniels, Daye Nam, Aayush Kumar, Stella Chen, Michael Hilton, Sruti Srinivasa Ragavan, and Mohammad Amin Alipour. 2024. Trust in Generative AI among Students: An exploratory study. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 1*. 67–73. https://doi.org/10.1145/3626252.3630842

[17] Matin Amoozadeh, Daye Nam, Daniel Prol, Ali Alfageeh, James Prather, Michael Hilton, Sruti Srinivasa Ragavan, and Amin Alipour. 2024. Student-ai interaction: A case study of CS1 students. In *Proceedings of the 24th Koli Calling International Conference on Computing Education Research*. 1–13. https://doi.org/10.1145/3699538.3699567

[18] Neil Anderson, Aidan McGowan, Philip Hanna, David Cutting, Leo Galway, and Matthew Collins. 2024. Using ChatGPT in Software Development Education. In *2024 IEEE Global Engineering Education Conference (EDUCON)*. IEEE, 1–5. https://doi.org/10.1109/EDUCON60312.2024.10578808

[19] Mikko Apiola, Henriikka Vartiainen, and Matti Tedre. 2024. First Year CS Students Exploring And Identifying Biases and Social Injustices in Text-to-Image Generative AI. In *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1*. 485–491. https://doi.org/10.1145/3649217.3653596

[20] Utkarsh Arora, Anupam Garg, Aryan Gupta, Samyak Jain, Ronit Mehta, Rupin Oberoi, Prachi, Aryaman Raina, Manav Saini, Sachin Sharma, et al. 2025. Analyzing llm usage in an advanced computing class in india. In *Proceedings of the 27th Australasian Computing Education Conference*. 154–163. https://doi.org/10.1145/3716640.3716657

[21] Andreas Axelsson, Daniel Tomas Wallgren, Udit Verma, Åsa Cajander, Mats Daniels, Anna Eckerdal, and Roger McDermott. 2024. From assistance to misconduct: Unpacking the complex role of generative ai in student learning. In *2024 IEEE Frontiers in Education Conference (FIE)*. IEEE, 1–8. https://doi.org/10.1109/FIE61694.2024.10893133

[22] Imen Azaiz, Natalie Kiesler, and Sven Strickroth. 2024. Feedback-generation for programming exercises with gpt-4. In *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1*. 31–37. https://doi.org/10.1145/3649217.3653594

[23] Rina Azoulay, Tirza Hirst, and Shulamit Reches. 2024. Integrity in the AI Era: Collaborative Solutions for Programming Education. In *International Conference on Agents and Artificial Intelligence*. Springer, 191–211. https://doi.org/10.1007/978-3-031-87330-0_10

[24] Rishabh Balse, Viraj Kumar, Prajish Prasad, and Jayakrishnan Madathil Warriem. 2023. Evaluating the quality of llm-generated explanations for logical errors in cs1 student programs. In *Proceedings of the 16th annual ACM India compute conference*. 49–54. https://doi.org/10.1145/3627217.3627233

[25] Jose Barambones, Cristian Moral, Angélica de Antonio, Ricardo Imbert, Lo¨ıc Martínez-Normand, and Elena Villalba-Mora. 2024. ChatGPT for learning HCI techniques: A case study on interviews for personas. *IEEE Transactions on Learning Technologies* 17 (2024), 1460–1475. https://doi.org/10.1109/TLT.2024.3386095

[26] Scott Barlowe, Daniel Aoulou, and Alex Ponce-Castillo. 2024. WIP: Generative AI as an Instructional Resource in a Computer Science Ethics Course. In *2024 IEEE Frontiers in Education Conference (FIE)*. IEEE, 1–5. https://doi.org/10.1109/FIE61694.2024.10892995

[27] Patrick Bassner, Eduard Frankford, and Stephan Krusche. 2024. Iris: An ai-driven virtual tutor for computer science education. In *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1*. 394–400. https://doi.org/10.1145/3649217.3653543

[28] Debarati Basu, Harini Ramaprasad, and Landon Nalewaja. 2024. Large-Scale Deployment and Evaluation of an Academic Integrity Module for Computing Students. In *2024 IEEE Frontiers in Education Conference (FIE)*. IEEE, 1–8. https://doi.org/10.1109/FIE61694.2024.10893045

[29] Brett A. Becker, Paul Denny, James Finnie-Ansley, Andrew Luxton-Reilly, James Prather, and Eddie Antonio Santos. 2023. Programming Is Hard - Or at Least It Used to Be: Educational Opportunities and Challenges of AI Code Generation. In *Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 1 (SIGCSE 2023)*. ACM, 500–506. https://doi.org/10.1145/3545945.3569759

[30] Seth Bernstein, Paul Denny, Juho Leinonen, Lauren Kan, Arto Hellas, Matt Littlefield, Sami Sarsa, and Stephen MacNeil. 2024. "Like a Nesting Doll": Analyzing Recursion Analogies Generated by CS Students Using Large Language Models. In *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1* (Milan, Italy) *(ITiCSE 2024)*. Association for Computing Machinery, New York, NY, USA, 122–128. https://doi.org/10.1145/3649217.3653533

[31] Mireilla Bikanga Ada. 2024. It Helps with Crap Lecturers and Their Low Effort: Investigating Computer Science Students' Perceptions of Using ChatGPT for Learning. *Education Sciences* 14, 10 (2024), 1106. https://doi.org/10.3390/educsci14101106

[32] William Bird. 2024. Faceless Adversary, Feckless Colleague: The Many Sides of ChatGPT. In *Proceedings of the 26th Western Canadian Conference on Computing Education*. 1–6. https://doi.org/10.1145/3660650.3660656

[33] Uwe M Borghoff, Mark Minas, and Jannis Schopp. 2025. Generative AI in Student Software Development Projects: A User Study on Experiences and Self-Assessment. In *Proceedings of the 6th European Conference on Software Engineering Education*. 161–170. https://doi.org/10.1145/3723010.3723012

[34] Dennis J Bouvier, Bruno Pereira Cipriano, Richard Glassey, Raymond Pettit, Emma Anderson, Anastasiia Birillo, Ryan Dougherty, Orit Hazzan, Olga Petrovska, Nuno Pombo, et al. 2025. GenAI Integration in Upper-Level Computing Courses. In *Proceedings of the 30th ACM Conference on Innovation and Technology in Computer Science Education V. 2*. 691–692. https://doi.org/10.1145/3724389.3731276

[35] Virginia Braun and Victoria Clarke. 2021. One size fits all? What counts as quality practice in (reflexive) thematic analysis? *Qualitative research in psychology* 18, 3 (2021), 328–352. https://doi.org/doi.org/10.1080/14780887.2020.1769238

[36] Allan Brockenbrough and Dominic Salinas. 2024. Using generative ai to create user stories in the software engineering classroom. In *2024 36th International Conference on Software Engineering Education and Training (CSEE&T)*. 1–5. https://doi.org/10.1109/CSEET62301.2024.10662994

[37] Ritvik Budhiraja, Ishika Joshi, Jagat Sesh Challa, Harshal D Akolekar, and Dhruv Kumar. 2024. "It's not like Jarvis, but it's pretty close!"-Examining ChatGPT's Usage among Undergraduate Students in Computer Science. In *Proceedings of the 26th Australasian Computing Education Conference*. 124–133. https://doi.org/10.1145/3636243.3636257

[38] Kevin Buffardi, JoAna Brooks, and David Alexander. 2024. Designing a CURE for CS1. In *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1*. 660–666. https://doi.org/10.1145/3649217.3653573

[39] Shubin Cai, Honglong Chen, Youyi Huang, and Zhong Ming. 2024. ComPAT: A Compiler Principles Course Assistant. In *International Conference on Knowledge Science, Engineering and Management*. Springer, 74–83. https://doi.org/10.1007/978-981-97-5489-2_7

[40] Zhenyao Cai, Seehee Park, Nia Nixon, and Shayan Doroudi. 2024. Advancing knowledge together: integrating large language model-based conversational AI in small group collaborative learning. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–9. https://doi.org/10.1145/3613905.3650868

[41] María Concepción Castillo-González and Agueda Ma Del Pilar Castillo-González. 2024. Ethical Challenges of AI: Nurturing Life-Care Perspectives in ECSIT Students. In *2024 IEEE International Symposium on Technology and Society (ISTAS)*. IEEE, 1–8. https://doi.org/10.1109/ISTAS61960.2024.10732170

[42] Devin Chamberlain, David B Levine, Abigail Pitcairn, Nicholas Snow, and Benjamin Sweeney. 2025. Large Language Models and Introductory Lab Exercises: Susceptibility, Resistance, and Potential. *Journal of Computing Sciences in Colleges* 40, 8 (2025), 49–63. https://doi.org/10.5555/3737313.3737332

[43] Binglin Chen, Colleen M Lewis, Matthew West, and Craig Zilles. 2024. Plagiarism in the age of generative ai: cheating method change and learning loss in an intro to CS course. In *Proceedings of the Eleventh ACM Conference on Learning@ Scale*. 75–85. https://doi.org/10.1145/3657604.3662046

[44] Sapna Cheryan, Victoria C Plaut, Caitlin Handron, and Lauren Hudson. 2013. The stereotypical computer scientist: Gendered media representations as a barrier to inclusion for women. *Sex roles* 69, 1 (2013), 58–71. https://doi.org/10.1007/s11199-013-0296-x

[45] Bruno Pereira Cipriano and Pedro Alves. 2023. GPT-3 vs Object Oriented Programming Assignments: An Experience Report. In *Proceedings of the 2023 Conference on Innovation and Technology in Computer Science Education V. 1* (Turku, Finland) *(ITiCSE 2023)*. Association for Computing Machinery, New York, NY, USA, 61–67. https://doi.org/10.1145/3587102.3588814

[46] Bruno Pereira Cipriano and Pedro Alves. 2024. LLMs still can't avoid instanceof: An investigation into GPT-3.5, GPT-4 and Bard's capacity to handle object-oriented programming assignments. In *Proceedings of the 46th International Conference on Software Engineering: Software Engineering Education and Training*. 162–169. https://doi.org/10.1145/3639474.3640052

[47] Bruno Pereira Cipriano, Pedro Alves, and Paul Denny. 2024. A picture is worth a thousand words: Exploring diagram and video-based oop exercises to counter llm over-reliance. In *European Conference on Technology Enhanced Learning*. Springer, 75–89. https://doi.org/10.1007/978-3-031-72315-5_6

[48] Aaron S Crandall, Bryan J Fischer, and Johannah L Crandall. 2024. WIP: ARTful Insights from a Pilot Study on GPT-Based Automatic Code Reviews in Undergraduate Computer Science Programs. In *2024 IEEE Frontiers in Education Conference (FIE)*. IEEE, 1–5. https://doi.org/10.1109/FIE61694.2024.10893407

[49] Aaron S. Crandall, Gina Sprint, and Bryan Fischer. 2023. Generative Pre Trained Transformer (GPT) Models as a Code Review Feedback Tool in Computer Science Programs. *J. Comput. Sci. Coll.* 39, 1 (Oct. 2023), 38–47. https://doi.org/10.5555/3636517.3636522

[50] Inés Friss de Kereki and Ismael Garrido. 2024. Solving Computer Science 2 Tasks: Students' Reflections on the Use of ChatGPT. In *2024 IEEE Global Engineering Education Conference (EDUCON)*. 1–5. https://doi.org/10.1109/EDUCON60312.2024.10578718

[51] Debzani Deb, Greg Taylor, Scott Betz, Bao Anh T Maddux, C Edward Ebert, Flourice W Richardson, Jeanine Lino S Couto, Michael S Jarrett, and Zagros Madjd-Sadjadi. 2025. Enhancing University Curricula with Integrated AI Ethics Education: A Comprehensive Approach. In *Proceedings of the 56th ACM Technical Symposium on Computer Science Education V. 1*. 248–254. https://doi.org/10.1145/3641554.3701953

[52] Andre del Carpio Gutierrez, Paul Denny, and Andrew Luxton-Reilly. 2024. Automating personalized parsons problems with customized contexts and concepts. In *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1*. 688–694. https://doi.org/10.1145/3649217.3653568

[53] Paul Denny, Juho Leinonen, James Prather, Andrew Luxton-Reilly, Thezyrie Amarouche, Brett A. Becker, and Brent N. Reeves. 2024. Prompt Problems: A New Programming Exercise for the Generative AI Era. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 1 (SIGCSE 2024)*. ACM, 296–302. https://doi.org/10.1145/3626252.3630909

[54] Paul Denny, Stephen MacNeil, Jaromir Savelka, Leo Porter, and Andrew Luxton-Reilly. 2024. Desirable Characteristics for AI Teaching Assistants in Programming Education. In *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1* (Milan, Italy) *(ITiCSE 2024)*. Association for Computing Machinery, New York, NY, USA, 408–414. https://doi.org/10.1145/3649217.3653574

[55] Paul Denny, Stephen MacNeil, Jaromir Savelka, Leo Porter, and Andrew Luxton-Reilly. 2024. Desirable Characteristics for AI Teaching Assistants in Programming Education. In *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1* (Milan, Italy) *(ITiCSE 2024)*. Association for Computing Machinery, New York, NY, USA, 408–414. https://doi.org/10.1145/3649217.3653574

[56] Paul Denny, James Prather, Brett A. Becker, James Finnie-Ansley, Arto Hellas, Juho Leinonen, Andrew Luxton-Reilly, Brent N. Reeves, Eddie Antonio Santos, and Sami Sarsa. 2024. Computing Education in the Era of Generative AI. *Commun. ACM* 67, 2 (Jan. 2024), 56–67. https://doi.org/10.1145/3624720

[57] Benjamin Denzler, Frank Vahid, Ashley Pang, and Mariam Salloum. 2024. Style anomalies can suggest cheating in CS1 programs. In *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1*. 381–387. https://doi.org/10.1145/3649217.3653626

[58] Daniel-Florin Dosaru, Diana-Maria Simion, Andrei-Horia Ignat, Lorina-Cristina Negreanu, and Alexandru-Corneliu Olteanu. 2024. A Code Analysis Tool to Help Students in the Age of Generative AI. In *European Conference on Technology Enhanced Learning*. Springer, 222–228. https://doi.org/10.1007/978-3-031-72312-4_31

[59] Jacob Doughty, Zipiao Wan, Anishka Bompelli, Jubahed Qayum, Taozhi Wang, Juran Zhang, Yujia Zheng, Aidan Doyle, Pragnya Sridhar, Arav Agarwal, Christopher Bogart, Eric Keylor, Can Kültür, Jaromír Savelka, and Majd Sakr. 2024. A Comparative Study of AI-Generated (GPT-4) and Human-Crafted MCQs in Programming Education. In *Proceedings of the 2024 Australian Computing Education Conference (ACE)*. ACM, Sydney, NSW, Australia, 114–123. https://doi.org/10.1145/3636243.3636256

[60] Nora Dunder, Saga Lundborg, Jacqueline Wong, and Olga Viberg. 2024. Kattis vs ChatGPT: Assessment and evaluation of programming tasks in the age of artificial intelligence. In *Proceedings of the 14th Learning Analytics and Knowledge Conference*. 821–827. https://doi.org/10.1145/3636555.3636882

[61] Ta Nguyen Binh Duong and Chai Yi Meng. 2024. Automatic grading of short answers using large language models in software engineering courses. In *2024 IEEE Global Engineering Education Conference (EDUCON)*. IEEE, 1–10. https://doi.org/10.1109/EDUCON60312.2024.10578839

[62] Jack Ebert and Kristina Kramarczuk. 2025. Leveraging Undergraduate Perspectives to Redefine AI Literacy. In *Proceedings of the 56th ACM Technical Symposium on Computer Science Education V. 1*. 290–296. https://doi.org/10.1145/3641554.3701916

[63] Aysa Fan, Ranran Haoran Zhang, Luc Paquette, and Rui Zhang. 2023. Exploring the Potential of Large Language Models in Generating Code-Tracing Questions for Introductory Programming Courses. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, Houda Bouamor, Juan Pino, and Kalika Bali (Eds.). Association for Computational Linguistics, Singapore, 7406–7421. https://doi.org/10.18653/v1/2023.findings-emnlp.496

[64] Laura Farinetti and Lorenzo Canale. 2024. Chatbot development using langchain: a case study to foster critical thinking and creativity. In *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1*. 401–407. https://doi.org/10.1145/3649217.3653557

[65] Michael Feffer, Nikolas Martelaro, and Hoda Heidari. 2023. The ai incident database as an educational tool to raise awareness of ai harms: A classroom exploration of efficacy, limitations, & future improvements. In *Proceedings of the 3rd ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization*. 1–11. https://doi.org/10.1145/3617694.3623223

[66] Molly Q Feldman and Carolyn Jane Anderson. 2024. Non-expert programmers in the generative AI future. In *Proceedings of the 3rd Annual Meeting of the Symposium on Human-Computer Interaction for Work*. 1–19. https://doi.org/10.1145/3663384.3663393

[67] Gianni Fenu, Roberta Galici, Mirko Marras, and Diego Reforgiato. 2024. Exploring student interactions with AI in programming training. In *Adjunct Proceedings of the 32nd ACM Conference on User Modeling, Adaptation and Personalization*. 555–560. https://doi.org/10.1145/3631700.3665227

[68] James Finnie-Ansley, Paul Denny, Brett A. Becker, Andrew Luxton-Reilly, and James Prather. 2022. The Robots Are Coming: Exploring the Implications of OpenAI Codex on Introductory Programming. In *Proceedings of the 24th Australasian Computing Education Conference (ACE '22)*. ACM, 10–19. https://doi.org/10.1145/3511861.3511863

[69] James Finnie-Ansley, Paul Denny, Andrew Luxton-Reilly, Eddie Antonio Santos, James Prather, and Brett A. Becker. 2023. My AI Wants to Know if This Will Be on the Exam: Testing OpenAI's Codex on CS2 Programming Exercises. In *Proceedings of the 25th Australasian Computing Education Conference*. ACM. https://doi.org/10.1145/3576123.3576134

[70] André Pimenta Freire, Paula Christina Figueira Cardoso, and André de Lima Salgado. 2023. May we consult chatgpt in our human-computer interaction written exam? an experience report after a professor answered yes. In *Proceedings of the XXII Brazilian Symposium on Human Factors in Computing Systems*. 1–11. https://doi.org/10.1145/3638067.3638100

[71] Hua Leong Fwa. 2024. Experience report: Identifying common misconceptions and errors of novice programmers with chatgpt. In *Proceedings of the 46th international conference on software engineering: Software engineering education and training*. 233–241. https://doi.org/10.1145/3639474.3640059

[72] Manuel B Garcia, Teodoro F Revano, Renato R Maaliw, Pitz Gerald G Lagrazon, Arlene Mae C Valderama, Ari Happonen, Basit Qureshi, and Ramazan Yilmaz. 2023. Exploring Student preference between aI-powered chatGPT and Human-curated stack overflow in resolving programming problems and queries. In *2023 IEEE 15th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM)*. IEEE, 1–6. https://doi.org/10.1109/HNICEM60674.2023.10589162

[73] Nicholas Gardella, Raymond Pettit, and Sara L Riggs. 2024. Performance, Workload, Emotion, and Self-Efficacy of Novice Programmers Using AI Code Generation. In *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1*. 290–296. https://doi.org/10.1145/3649217.3653615

[74] Edward F. Gehringer, Jianxun George Wang, and Sharan Kumar Jilla. 2024. Dual-Submission Homework in Parallel Computer Architecture: An Exploratory Study in the Age of LLMs. In *Proceedings of the Workshop on Computer Architecture Education* (Orlando, FL, USA) *(WCAE '23)*. Association for Computing Machinery, New York, NY, USA, 41–47. https://doi.org/10.1145/3605507.3610629

[75] Isaac Geller, Daniel Liss, Brett A. Becker, and Shriram Krishnamurthi. 2024. Understanding Student-LLM Interactions: A Case Study Using GPT-4. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education (SIGCSE)*. ACM, 958–964. https://doi.org/10.1145/3641554.3701800

[76] Nasser Giacaman and Valerio Terragni. 2025. Empowering Computing Students with Large Language Models by Developing an Escape Room Game. 6 pages. https://doi.org/10.1145/3708897

[77] Virginia Grande, Natalie Kiesler, and María Andreína Francisco R. 2024. Student Perspectives on Using a Large Language Model (LLM) for an Assignment on Professional Ethics. In *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1* (Milan, Italy) *(ITiCSE 2024)*. Association for Computing Machinery, New York, NY, USA, 478–484. https://doi.org/10.1145/3649217.3653624

[78] Skyler Grandel, Douglas C. Schmidt, and Kevin Leach. 2024. Applying Large Language Models to Enhance the Assessment of Parallel Functional Programming Assignments. In *Proceedings of the 1st International Workshop on Large Language Models for Code* (Lisbon, Portugal) *(LLM4Code '24)*. Association for Computing Machinery, New York, NY, USA, 102–110. https://doi.org/10.1145/3643795.3648375

[79] Christian Grévisse, Maria Angeliki S Pavlou, and Jochen G Schneider. 2024. Docimological quality analysis of LLM-generated multiple choice questions in computer science and medicine. *SN Computer Science* 5, 5 (2024), 636. https://doi.org/0.1007/s42979-024-02963-6

[80] Sebastian Gutierrez, Irene Hou, Jihye Lee, Kenneth Angelikas, Owen Man, Sophia Mettille, James Prather, Paul Denny, and Stephen MacNeil. 2025. Seeing the Forest and the Trees: Solving Visual Graph and Tree Based Data Structure Problems using Large Multimodal Models. In *Proceedings of the 27th Australasian Computing Education Conference (ACE '25)*. Association for Computing Machinery, New York, NY, USA, 124–133. https://doi.org/10.1145/3716640.3716654

[81] Aldric Gutiérrez-Ferré, Davinia Hernández-Leo, and J Roberto Sánchez-Reina. 2024. Generative AI chatbot in PyramidApp: students' behaviors and design principles. In *International Conference on Collaboration Technologies and Social Computing*. Springer, 248–256. https://doi.org/10.1007/978-3-031-67998-8_19

[82] Philipp Haindl and Gerald Weinberger. 2024. Students' Experiences of Using ChatGPT in an Undergraduate Programming Course. *IEEE Access* 12 (2024), 43519–43529. https://doi.org/10.1109/ACCESS.2024.3380909

[83] Olaf Hallan Graven and Lachlan MacKinnon. 2023. Developing Higher Education – Post-Pandemic – Influenced by AI. In *2023 IEEE Frontiers in Education Conference (FIE)*. 01–08. https://doi.org/10.1109/FIE58773.2023.10343215

[84] Sabine Hammer, Sarah Ottinger, Benedikt Zönnchen, Michel Hohendanner, Martin Hobelsberger, and Veronika Thurner. 2024. ChatGPT in Higher Education: Perceptions of Computer Science-Related Students. In *2024 IEEE Global Engineering Education Conference (EDUCON)*. 01–08. https://doi.org/10.1109/EDUCON60312.2024.10578742

[85] Emily Hammond and Courtney Faber. 2024. Work in Progress: Integration of AI Tools on an Open-Ended Computer Programming Project. In *2024 IEEE Frontiers in Education Conference (FIE)*. 1–5. https://doi.org/10.1109/FIE61694.2024.10892956

[86] Alisha HASAN, Bogdan SIMION, and Florin POP. 2024. Analyzing the uses and perceptions of computer science students towards generative AI tools. In *ON VIRTUAL LEARNING*. 363. https://doi.org/10.58503/icvl-v19y202430

[87] Victor Takashi Hayashi and Henrique Mohallem Paiva. 2024. ChatGPT Calls for Self Reflection: Student Perceptions of Evaluation Activities in Video. In *2024 IEEE International Conference on Teaching, Assessment and Learning for Engineering (TALE)*. 1–5. https://doi.org/10.1109/TALE62452.2024.10834346

[88] Zhangying He, Thomas Nguyen, Tahereh Miari, Mehrdad Aliasgari, Setareh Rafatirad, and Hossein Sayadi. 2024. The AI Companion in Education: Analyzing the Pedagogical Potential of ChatGPT in Computer Science and Engineering. In *2024 IEEE Global Engineering Education Conference (EDUCON)*. 1–10. https://doi.org/10.1109/EDUCON60312.2024.10578820

[89] Malcolm Yoke Hean Low, Chien Ching Lee, Keok Kee Lee, and Kuet Leong Lam. 2024. Enhancing the Teaching of Data Structures and Algorithms using AI Chatbots. In *2024 IEEE International Conference on Teaching, Assessment and Learning for Engineering (TALE)*. 1–8. https://doi.org/10.1109/TALE62452.2024.10834293

[90] Arto Hellas, Juho Leinonen, Sami Sarsa, Charles Koutcheme, Lilja Kujanp¨a¨a, and Juha Sorva. 2023. Exploring the Responses of Large Language Models to Beginner Programmers' Help Requests. In *Proceedings of the 2023 ACM Conference on International Computing Education Research - Volume 1* (Chicago, IL, USA) *(ICER '23)*. Association for Computing Machinery, New York, NY, USA, 93–105. https://doi.org/10.1145/3568813.3600139

[91] Muntasir Hoq, Yang Shi, Juho Leinonen, Damilola Babalola, Collin F Lynch, and Bita Akram. 2023. Detecting ChatGPT-Generated Code in a CS1 Course.. In *LLM@ AIED*. 53–63.

[92] Md Abu Hossain, Md Rafid Hasan, Mohammad Tauhidur Rahman, and Amr Mohamed. 2024. Diffusion Models in Natural Language Processing: A Comprehensive Survey. In *2024 International Conference on Computer, Information and Telecommunication Systems (CITS)*. IEEE, 1–6. https://doi.org/10.1109/CITS59986.2024.10546497

[93] Irene Hou, Owen Man, Kate Hamilton, Srishty Muthusekaran, Jeffin Johnykutty, Leili Zadeh, and Stephen MacNeil. 2025. 'All Roads Lead to ChatGPT': How Generative AI is Eroding Social Interactions and Student Learning Communities. In *Proceedings of the 30th ACM Conference on Innovation and Technology in Computer Science Education V. 1* (Nijmegen, Netherlands) *(ITiCSE 2025)*. Association for Computing Machinery, New York, NY, USA, 79–85. https://doi.org/10.1145/3724363.3729024

[94] Irene Hou, Owen Man, Sophia Mettille, Sebastian Gutierrez, Kenneth Angelikas, and Stephen MacNeil. 2024. More Robots are Coming: Large Multimodal Models (ChatGPT) can Solve Visually Diverse Images of Parsons Problems. In *Proceedings of the 26th Australasian Computing Education Conference* (Sydney, NSW, Australia) *(ACE '24)*. Association for Computing Machinery, New York, NY, USA, 29–38. https://doi.org/10.1145/3636243.3636247

[95] Irene Hou, Sophia Mettille, Owen Man, Zhuo Li, Cynthia Zastudil, and Stephen MacNeil. 2024. The Effects of Generative AI on Computing Students' Help-Seeking Preferences. In *Proceedings of the 26th Australasian Computing Education Conference* (Sydney, NSW, Australia) *(ACE '24)*. Association for Computing Machinery, New York, NY, USA, 39–48. https://doi.org/10.1145/3636243.3636248

[96] Wen-Jung Hsin. 2024. The Effect of ChatGPT: Student Perspective and Performance Achievement. *Journal of Computing Sciences in Colleges* 39, 6 (2024), 20–29. https://doi.org/10.5555/3665464.3665467

[97] Minjie Hu, Tony Assadi, and Hamid Mahroeian. 2023. Explicitly Introducing ChatGPT into First-year Programming Practice: Challenges and Impact. In *2023 IEEE International Conference on Teaching, Assessment and Learning for Engineering (TALE)*. 1–6. https://doi.org/10.1109/TALE56641.2023.10398297

[98] Niklas Humble, Jonas Boustedt, Hanna Holmgren, Goran Milutinovic, Stefan Seipel, and Ann-Sofie ¨Ostberg. 2023. Cheaters or AI-enhanced learners: Consequences of ChatGPT for programming education. *Electronic journal of e-Learning* 22, 2 (2023), 16–29. https://doi.org/10.34190/ejel.21.5.3154

[99] Shereen Ismail, Jenice Lynch, and Runna Alghazo. 2024. Enhancing STEM Education with ChatGPT: A Case Study on Developing Practice Assessments for C Programming. In *2024 IEEE 13th International Conference on Engineering Education (ICEED)*. 1–6. https://doi.org/10.1109/ICEED62316.2024.10923780

[100] Sven Jacobs and Steffen Jaschke. 2024. Evaluating the Application of Large Language Models to Generate Feedback in Programming Education. In *2024 IEEE Global Engineering Education Conference (EDUCON)*. 1–5. https://doi.org/10.1109/EDUCON60312.2024.10578838

[101] Sven Jacobs and Steffen Jaschke. 2024. Leveraging Lecture Content for Improved Feedback: Explorations with GPT-4 and Retrieval Augmented Generation. In *2024 36th International Conference on Software Engineering Education and Training (CSEE&T)*. 1–5. https://doi.org/10.1109/CSEET62301.2024.10663001

[102] Pooriya Jamie, Reyhaneh HajiHashemi, and Sharareh Alipour. 2025. Utilizing ChatGPT in a Data Structures and Algorithms Course: A Teaching Assistant's Perspective. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*. ACM, 1–7. https://doi.org/10.1145/3706599.3720291

[103] Peter Jamieson, Suman Bhunia, and Dhananjai M. Rao. 2023. With ChatGPT, Do We have to Rewrite Our Learning Objectives - CASE Study in Cybersecurity. In *2023 IEEE Frontiers in Education Conference (FIE)*. 1–5. https://doi.org/10.1109/FIE58773.2023.10343052

[104] Hyoungwook Jin, Seonghee Lee, Hyungyu Shin, and Juho Kim. 2024. Teach AI How to Code: Using Large Language Models as Teachable Agents for Programming Education. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) *(CHI '24)*. Association for Computing Machinery, New York, NY, USA, Article 652, 28 pages. https://doi.org/10.1145/3613904.3642349

[105] Mollie Jordan, Kevin Ly, and Adalbert Gerald Soosai Raj. 2024. Need a Programming Exercise Generated in Your Native Language? ChatGPT's Got Your Back: Automatic Generation of Non-English Programming Exercises Using OpenAI GPT-3.5. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 1* (Portland, OR, USA) *(SIGCSE 2024)*. Association for Computing Machinery, New York, NY, USA, 618–624. https://doi.org/10.1145/3626252.3630897

[106] Ishika Joshi, Ritvik Budhiraja, Harshal Dev, Jahnvi Kadia, Mohammad Osama Ataullah, Sayan Mitra, Harshal D. Akolekar, and Dhruv Kumar. 2024. ChatGPT in the Classroom: An Analysis of Its Strengths and Weaknesses for Solving Undergraduate Computer Science Questions. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 1* (Portland, OR, USA) *(SIGCSE 2024)*. Association for Computing Machinery, New York, NY, USA, 625–631. https://doi.org/10.1145/3626252.3630803

[107] Breanna Jury, Angela Lorusso, Juho Leinonen, Paul Denny, and Andrew Luxton-Reilly. 2024. Evaluating LLM-generated Worked Examples in an Introductory Programming Course. In *Proceedings of the 26th Australasian Computing Education Conference* (Sydney, NSW, Australia) *(ACE '24)*. Association for Computing Machinery, New York, NY, USA, 77–86. https://doi.org/10.1145/3636243.3636252

[108] FNU Kaleemunnisa, Christelle Scharff, Krishna Bathula, and Begimai Zhumakova. 2024. ChatGPT in the Classroom: Experimentation in a Python Class for Non-Computing Majors. In *2024 IEEE Integrated STEM Education Conference (ISEC)*. 1–7. https://doi.org/10.1109/ISEC61299.2024.10665172

[109] Amanpreet Kapoor, Marc Diaz, Stephen MacNeil, Leo Porter, and Paul Denny. 2025. Exploring Student Behaviors and Motivations using AI TAs with Optional

Guardrails. *arXiv preprint arXiv:2504.11146* (2025).

[110] Oscar Karnalim, Hapnes Toba, and Meliana Christianti Johan. 2024. Detecting AI assisted submissions in introductory programming via code anomaly. *Education and Information Technologies* 29, 13 (2024), 16841–16866. https://doi.org/10.1007/s10639-024-12520-6

[111] Oscar Karnalim, Hapnes Toba, Meliana Christianti Johan, Erico Darmawan Handoyo, Yehezkiel David Setiawan, and Josephine Alvina Luwia. 2023. Plagiarism and AI Assistance Misuse in Web Programming: Unfair Benefits and Characteristics. In *2023 IEEE International Conference on Teaching, Assessment and Learning for Engineering (TALE)*. 1–5. https://doi.org/10.1109/TALE56641.2023.10398397

[112] Majeed Kazemitabaar, Runlong Ye, Xiaoning Wang, Austin Zachary Henley, Paul Denny, Michelle Craig, and Tovi Grossman. 2024. Codeaid: Evaluating a classroom deployment of an llm-based programming assistant that balances student and educator needs. In *Proceedings of the 2024 chi conference on human factors in computing systems*. 1–20. https://doi.org/10.1145/3613904.3642773

[113] Chris Kerslake, Paul Denny, David H. Smith, James Prather, Juho Leinonen, Andrew Luxton-Reilly, and Stephen MacNeil. 2024. Integrating Natural Language Prompting Tasks in Introductory Programming Courses. In *Proceedings of the 2024 on ACM Virtual Global Computing Education Conference V. 1* (Virtual Event, NC, USA) *(SIGCSE Virtual 2024)*. Association for Computing Machinery, New York, NY, USA, 88–94. https://doi.org/10.1145/3649165.3690125

[114] Hieke Keuning, Isaac Alpizar-Chacon, Ioanna Lykourentzou, Lauren Beehler, Christian K¨oppe, Imke de Jong, and Sergey Sosnovsky. 2024. Students' Perceptions and Use of Generative AI Tools for Programming Across Different Computing Courses. In *Proceedings of the 24th Koli Calling International Conference on Computing Education Research (Koli Calling '24)*. Association for Computing Machinery, New York, NY, USA, Article 14, 12 pages. https://doi.org/10.1145/3699538.3699546

[115] Natalie Kiesler, Dominic Lohr, and Hieke Keuning. 2023. Exploring the Potential of Large Language Models to Generate Formative Programming Feedback. In *2023 IEEE Frontiers in Education Conference (FIE)*. 1–5. https://doi.org/10.1109/FIE58773.2023.10343457

[116] Bailey Kimmel, Austin Lee Geisert, Lily Yaro, Brendan Gipson, Ronald Taylor Hotchkiss, Sidney Kwame Osae-Asante, Hunter Vaught, Grant Wininger, and Chase Yamaguchi. 2024. Enhancing Programming Error Messages in Real Time with Generative AI. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) *(CHI EA '24)*. Association for Computing Machinery, New York, NY, USA, Article 608, 7 pages. https://doi.org/10.1145/3613905.3647967

[117] Harsh Kumar, Ilya Musabirov, Mohi Reza, Jiakai Shi, Xinyuan Wang, Joseph Jay Williams, Anastasia Kuzminykh, and Michael Liut. 2024. Guiding Students in Using LLMs in Supported Learning Environments: Effects on Interaction Dynamics, Learner Performance, Confidence, and Trust. *Proceedings of the ACM on Human-Computer Interaction* 8, CSCW2 (2024), 1–30. https://doi.org/10.1145/3687038

[118] Yulia Kumar, Anjana Manikandan, J. Jenny Li, and Patricia Morreale. 2024. Optimizing Large Language Models for Auto-Generation of Programming Quizzes. In *2024 IEEE Integrated STEM Education Conference (ISEC)*. 1–5. https://doi.org/10.1109/ISEC61299.2024.10665141

[119] Sam Lau and Philip Guo. 2023. From "Ban It Till We Understand It" to "Resistance is Futile": How University Programming Instructors Plan to Adapt as More Students Use AI Code Generation and Explanation Tools such as ChatGPT and GitHub Copilot. In *Proceedings of the 2023 ACM Conference on International Computing Education Research - Volume 1 (ICER '23)*. ACM, 106–121. https://doi.org/10.1145/3568813.3600138

[120] Sam Lau and Philip Guo. 2023. From ""Ban It Till We Understand It"" to ""Resistance is Futile"": How University Programming Instructors Plan to Adapt as More Students Use AI Code Generation and Explanation Tools such as ChatGPT and GitHub Copilot. In *Proceedings of the 2023 ACM Conference on International Computing Education Research - Volume 1* (Chicago, IL, USA) *(ICER '23)*. Association for Computing Machinery, New York, NY, USA, 106–121. https://doi.org/10.1145/3568813.3600138

[121] Teemu Lehtinen, Charles Koutcheme, and Arto Hellas. 2024. Let's Ask AI About Their Programs: Exploring ChatGPT's Answers To Program Comprehension Questions. In *Proceedings of the 46th International Conference on Software Engineering: Software Engineering Education and Training* (Lisbon, Portugal) *(ICSE-SEET '24)*. Association for Computing Machinery, New York, NY, USA, 221–232. https://doi.org/10.1145/3639474.3640058

[122] Juho Leinonen, Paul Denny, Stephen MacNeil, Sami Sarsa, Seth Bernstein, Joanne Kim, Andrew Tran, and Arto Hellas. 2023. Comparing Code Explanations Created by Students and Large Language Models. In *Proceedings of the 2023 Conference on Innovation and Technology in Computer Science Education V. 1* (Turku, Finland) *(ITiCSE 2023)*. Association for Computing Machinery, New York, NY, USA, 124–130. https://doi.org/10.1145/3587102.3588785

[123] Juho Leinonen, Arto Hellas, Sami Sarsa, Brent Reeves, Paul Denny, James Prather, and Brett A. Becker. 2023. Using Large Language Models to Enhance Programming Error Messages. In *Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 1* (Toronto ON, Canada) *(SIGCSE 2023)*. Association for Computing Machinery, New York, NY, USA, 563–569. https://doi.org/10.1145/3545945.3569770

[124] Karol Lejmbach and Sean Mackay. 2024. Using Chat-GPT to Create Multiple Choice CS Exams. In *2024 IEEE Frontiers in Education Conference (FIE)*. 1–5. https://doi.org/10.1109/FIE61694.2024.10893096

[125] Yi Li, Riteng Zhang, Danni Qu, and Maíra Marques Samary. 2023. Leveraging LLMs and MLPs in Designing a Computer Science Placement Test System. In *2023 International Conference on Computational Science and Computational Intelligence (CSCI)*. 1670–1676. https://doi.org/10.1109/CSCI62032.2023.00276

[126] Mark Liffiton, Brad E Sheese, Jaromir Savelka, and Paul Denny. 2024. CodeHelp: Using Large Language Models with Guardrails for Scalable Support in Programming Classes. In *Proceedings of the 23rd Koli Calling International Conference on Computing Education Research* (Koli, Finland) *(Koli Calling '23)*. Association for Computing Machinery, New York, NY, USA, Article 8, 11 pages. https://doi.org/10.1145/3631802.3631830

[127] Suqing Liu, Zezhu Yu, Feiran Huang, Yousef Bulbulia, Andreas Bergen, and Michael Liut. 2024. Can Small Language Models With Retrieval-Augmented Generation Replace Large Language Models When Learning Computer Science?. In *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1* (Milan, Italy) *(ITiCSE 2024)*. Association for Computing Machinery, New York, NY, USA, 388–393. https://doi.org/10.1145/3649217.3653554

[128] Yangtao Liu, Hengyuan Liu, Zezhong Yang, Zheng Li, and Yong Liu. 2024. Empirical Evaluation of Large Language Models for Novice Program Fault Localization. In *2024 IEEE 24th International Conference on Software Quality, Reliability and Security (QRS)*. 180–191. https://doi.org/10.1109/QRS62785.2024.00027

[129] Evanfiya Logacheva, Arto Hellas, James Prather, Sami Sarsa, and Juho Leinonen. 2024. Evaluating Contextually Personalized Programming Exercises Created with Generative AI. In *Proceedings of the 2024 ACM Conference on International Computing Education Research - Volume 1* (Melbourne, VIC, Australia) *(ICER '24)*. Association for Computing Machinery, New York, NY, USA, 95–113. https://doi.org/10.1145/3632620.3671103

[130] Robert Logozar, Matija Mikac, and Jurica Hizak. 2023. ChatGPT on the Freshman Test in C/C++ Programming. In *2023 IEEE 21st Jubilee International Symposium on Intelligent Systems and Informatics (SISY)*. 000255–000264. https://doi.org/10.1109/SISY60376.2023.10417871

[131] Dominic Lohr, Hieke Keuning, and Natalie Kiesler. 2025. You're (Not) My Type-Can LLMs Generate Feedback of Specific Types for Introductory Programming Tasks? *Journal of Computer Assisted Learning* 41, 1 (2025), e13107. https://doi.org/10.1111/jcal.13107

[132] Wenhan Lyu, Yimeng Wang, Tingting (Rachel) Chung, Yifan Sun, and Yixuan Zhang. 2024. Evaluating the Effectiveness of LLMs in Introductory Computer Science Education: A Semester-Long Field Study. In *Proceedings of the Eleventh ACM Conference on Learning @ Scale* (Atlanta, GA, USA) *(L@S '24)*. Association for Computing Machinery, New York, NY, USA, 63–74. https://doi.org/10.1145/3657604.3662036

[133] Boxuan Ma, Li Chen, and Shin'ichi Konomi. 2024. Enhancing programming education with ChatGPT: A case study on student perceptions and interactions in a Python course. In *International Conference on Artificial Intelligence in Education*. Springer, 113–126. https://doi.org/10.1007/978-3-031-64315-6_9

[134] Boxuan Ma, Li Chen, and Shin'ichi Konomi. 2024. Exploring Student Perception and Interaction Using ChatGPT in Programming Education. *International Association for Development of the Information Society* (2024).

[135] Kelly Mack, Emma McDonnell, Dhruv Jain, Lucy Lu Wang, Jon E. Froehlich, and Leah Findlater. 2021. What do we mean by "accessibility research"? A literature survey of accessibility papers in CHI and ASSETS from 1994 to 2019. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. https://doi.org/10.1145/3411764.3445412

[136] Stephen MacNeil, Paul Denny, Andrew Tran, Juho Leinonen, Seth Bernstein, Arto Hellas, Sami Sarsa, and Joanne Kim. 2024. Decoding Logic Errors: A Comparative Study on Bug Detection by Students and Large Language Models. In *Proceedings of the 26th Australasian Computing Education Conference* (Sydney, NSW, Australia) *(ACE '24)*. Association for Computing Machinery, New York, NY, USA, 11–18. https://doi.org/10.1145/3636243.3636245

[137] Stephen MacNeil, James Prather, Rahad Arman Nabid, Sebastian Gutierrez, Silas Carvalho, Saimon Shrestha, Paul Denny, Brent N Reeves, Juho Leinonen, and Rachel Louise Rossetti. 2025. Fostering Responsible AI Use Through Negative Expertise: A Contextualized Autocompletion Quiz. In *Proceedings of the 30th ACM Conference on Innovation and Technology in Computer Science Education V. 1*. 326–332. https://doi.org/10.1145/3724363.3729067

[138] Stephen MacNeil, Andrew Tran, Arto Hellas, Joanne Kim, Sami Sarsa, Paul Denny, Seth Bernstein, and Juho Leinonen. 2023. Experiences from Using Code Explanations Generated by Large Language Models in a Web Software Development E-Book. In *Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 1* (Toronto ON, Canada) *(SIGCSE 2023)*. Association for Computing Machinery, New York, NY, USA, 931–937. https://doi.org/10.1145/3545945.3569785

[139] Alina Mailach, Dominik Gorgosch, Norbert Siegmund, and Janet Siegmund. 2025. "Ok Pal, we have to code that now": interaction patterns of programming beginners with a conversational chatbot. *Empirical Software Engineering* 30, 1 (2025), 34. https://doi.org/0.1007/s10664-024-10561-6

[140] Eric D. Manley, Timothy Urness, Andrei Migunov, and Md. Alimoor Reza. 2024. Examining Student Use of AI in CS1 and CS2. *J. Comput. Sci. Coll.* 39, 6 (April 2024), 41–51. https://doi.org/10.5555/3665464.3665469

[141] Sathiamoorthy Manoharan, Ulrich Speidel, Anthony Edward Ward, and Xinfeng Ye. 2023. Contract Cheating – Dead or Reborn?. In *2023 32nd Annual Conference of the European Association for Education in Electrical and Information Engineering (EAEEIE)*. 1–5. https://doi.org/10.23919/EAEEIE55804.2023.10182073

[142] Lauren E. Margulieux, James Prather, Brent N. Reeves, Brett A. Becker, Gozde Cetin Uzun, et al. 2024. Self-Regulation, Self-Efficacy, and Fear of Failure Interactions with How Novices Use LLMs to Solve Programming Problems. In *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1 (ITiCSE 2024)*. ACM, 276–282. https://doi.org/10.1145/3649217.3653621

[143] John Ivan Curbano Maurat, Elsie Villareal Isip, and Aileen Gail Regala Lumabas. 2025. A Comparative Study of Gender Differences in the Utilization and Effectiveness of AI-Assisted Learning Tools in Programming Among University Students. In *Proceedings of the 2024 International Conference on Artificial Intelligence and Teacher Education (ICAITE '24)*. Association for Computing Machinery, New York, NY, USA, 30–34. https://doi.org/10.1145/3702386.3702392

[144] Bradley McDanel and Ed Novak. 2025. Designing LLM-Resistant Programming Assignments: Insights and Strategies for CS Educators. In *Proceedings of the 56th ACM Technical Symposium on Computer Science Education V. 1* (Pittsburgh, PA, USA) *(SIGCSETS 2025)*. Association for Computing Machinery, New York, NY, USA, 756–762. https://doi.org/10.1145/3641554.3701872

[145] Tyler Menezes, Lola Egherman, and Nikhil Garg. 2024. AI-Grading Standup Updates to Improve Project-Based Learning Outcomes. In *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1* (Milan, Italy) *(ITiCSE 2024)*. Association for Computing Machinery, New York, NY, USA, 17–23. https://doi.org/10.1145/3649217.3653541

[146] Regina Merine and Saptarshi Purkayastha. 2022. Risks and Benefits of AI-generated Text Summarization for Expert Level Content in Graduate Health Informatics. In *2022 IEEE 10th International Conference on Healthcare Informatics (ICHI)*. 567–574. https://doi.org/10.1109/ICHI54592.2022.00113

[147] Federico Meza, Oscar Acevedo, and Matías González. 2024. Improving the Efficacy of an Automated Judging System in an Introductory Programming Class using a Large Language Model. In *2024 IEEE 42nd Central America and Panama Convention (CONCAPAN XLII)*. IEEE, 1–6. https://doi.org/10.1109/CONCAPAN63470.2024.10933867

[148] Simone Mezzaro, Alessio Gambi, and Gordon Fraser. 2024. An Empirical Study on How Large Language Models Impact Software Testing Learning. In *Proceedings of the 28th International Conference on Evaluation and Assessment in Software Engineering* (Salerno, Italy) *(EASE '24)*. Association for Computing Machinery, New York, NY, USA, 555–564. https://doi.org/10.1145/3661167.3661273

[149] Diego Miranda, Dayana Palma, Adrian Fernández, René Noel, Cristian Cechinel, and Roberto Munoz. 2024. Enhancing Agile Project Management Education with AI: ChatGPT-4's Role in Evaluating Student Contributions. In *2024 43rd International Conference of the Chilean Computer Science Society (SCCC)*. IEEE, 1–4. https://doi.org/10.1109/SCCC63879.2024.10767612

[150] Horia Alexandru Modran, Doru Ursuțiu, Cornel Samoilă, and Elena-Cătălina Gherman-Dolhăscu. 2024. Developing a GPT Chatbot Model for Students Programming Education. In *International Conference on Smart Technologies & Education*. Springer, 72–82. https://doi.org/10.1007/978-3-031-61905-2_8

[151] Kareem Mohamed, Mina Yousef, Walaa Medhat, Ensaf Hussein Mohamed, Ghada Khoriba, and Tamer Arafa. 2025. Hands-on analysis of using large language models for the auto evaluation of programming assignments. *Information Systems* 128 (2025), 102473. https://doi.org/10.1016/j.is.2024.102473

[152] Mohamed Morsy, Abdallah Farraj, and David Reavis. 2023. On the Challenges and Opportunities of Using ChatGPT in Academia. In *2023 IEEE Frontiers in Education Conference (FIE)*. 01–06. https://doi.org/10.1109/FIE58773.2023.10343189

[153] Ritwik Murali, Dhanya M. Dhanalakshmy, Veeramanohar Avudaiappan, and Gayathri Sivakumar. 2024. Towards Assessing the Credibility of Chatbot Responses for Technical Assessments in Higher Education. In *2024 IEEE Global Engineering Education Conference (EDUCON)*. 1–9. https://doi.org/10.1109/EDUCON60312.2024.10578934

[154] Ritwik Murali, Nitin Ravi, and Amruthiyu Surendran. 2024. Augmenting Virtual Labs with Artificial Intelligence for Hybrid Learning. In *2024 IEEE Global Engineering Education Conference (EDUCON)*. 1–10. https://doi.org/10.1109/EDUCON60312.2024.10578649

[155] Radka Nacheva and Iviana Hristova. 2024. AI Chatbots for Competency-Based Training in Programming Languages Courses. In *2024 International Conference on Intelligent Computing and Next Generation Networks (ICNGN)*. 01–05. https://doi.org/10.1109/ICNGN63705.2024.10871836

[156] Atharva Naik, Jessica Ruhan Yin, Anusha Kamath, Qianou Ma, Sherry Tongshuang Wu, Charles Murray, Christopher Bogart, Majd Sakr, Carolyn Penstein Rose, Andrew M. Olney, Irene-Angelica Chounta, Zitao Liu, Olga C. Santos, and Ig Ibert Bittencourt. 2024. Generating Situated Reflection Triggers About Alternative Solution Paths: A Case Study of Generative AI for Computer-Supported Collaborative Learning. In *Artificial Intelligence in Education*. Springer Nature Switzerland, Cham, 46–59. https://doi.org/10.1007/978-3-031-64302-6_4

[157] Sagnik Nath and So Yoon Yoon. 2024. WIP: Beyond Code: Evaluating ChatGPT, Gemini, Claude, and Meta AI as AI Tutors in Computer Science and Engineering Education. In *2024 IEEE Frontiers in Education Conference (FIE)*. 1–5. https://doi.org/10.1109/FIE61694.2024.10893528

[158] Alexander Tobias Neumann, Yue Yin, Sulayman Sowe, Stefan Decker, and Matthias Jarke. 2025. An LLM-Driven Chatbot in Higher Education for Databases and Information Systems. *IEEE Transactions on Education* 68, 1 (2025), 103–116. https://doi.org/10.1109/TE.2024.3467912

[159] Andres Neyem, Juan Pablo Sandoval Alcocer, Marcelo Mendoza, Leonardo Centellas-Claros, Luis A. Gonzalez, and Carlos Paredes-Robles. 2024. Exploring the Impact of Generative AI for StandUp Report Recommendations in Software Capstone Project Development. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 1* (Portland, OR, USA) *(SIGCSE 2024)*. Association for Computing Machinery, New York, NY, USA, 951–957. https://doi.org/10.1145/3626252.3630854

[160] Ha Nguyen and Vicki Allan. 2024. Using GPT-4 to Provide Tiered, Formative Code Feedback. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 1* (Portland, OR, USA) *(SIGCSE 2024)*. Association for Computing Machinery, New York, NY, USA, 958–964. https://doi.org/10.1145/3626252.3630960

[161] Ha Nguyen, Nate Stott, and Vicki Allan. 2024. Comparing Feedback from Large Language Models and Instructors: Teaching Computer Science at Scale. In *Proceedings of the Eleventh ACM Conference on Learning @ Scale* (Atlanta, GA, USA) *(L@S '24)*. Association for Computing Machinery, New York, NY, USA, 335–339. https://doi.org/10.1145/3657604.3664660

[162] Faisal Nizamudeen, Lorenzo Gatti, Nacir Bouali, and Faizan Ahmed. 2024. Investigating the impact of code generation tools (chatgpt & github copilot) on programming education. In *16th International Conference on Computer Supported Education, CSEDU 2024*. Science and Technology Publications, Lda, 221–229. https://doi.org/10.5220/0012628000003693

[163] Harsha Nutalapati, Sachin Velmurugan, and Nestor Michael Tiglao. 2024. Coding Buddy: An Adaptive AI-Powered Platform for Personalized Learning. In *2024 International Symposium on Networks, Computers and Communications (ISNCC)*. 1–6. https://doi.org/10.1109/ISNCC62547.2024.10759044

[164] Sunggyeol Oh, Yi Cao, Andrew Katz, and Jialu Zhao. 2024. Explore Public's Perspectives on Generative AI in Computer Science (CS) Education: A Social Media Data Analysis. In *2024 IEEE Frontiers in Education Conference (FIE)*. 1–9. https://doi.org/10.1109/FIE61694.2024.10893102

[165] Sanghak Oh, Kiho Lee, Seonhye Park, Doowon Kim, and Hyoungshick Kim. 2024. Poisoned ChatGPT Finds Work for Idle Hands: Exploring Developers' Coding Practices with Insecure Suggestions from Poisoned AI Models. In *2024 IEEE Symposium on Security and Privacy (SP)*. 1141–1159. https://doi.org/10.1109/SP54263.2024.00046

[166] Fan Ouyang, Mingyue Guo, Ning Zhang, Xianping Bai, and Pengcheng Jiao. 2024. Comparing the Effects of Instructor Manual Feedback and ChatGPT Intelligent Feedback on Collaborative Programming in China's Higher Education. *IEEE Transactions on Learning Technologies* 17 (2024), 2173–2185. https://doi.org/10.1109/TLT.2024.3486749

[167] Solomon Sunday Oyelere and Kehinde Aruleba. 2025. A comparative study of student perceptions on generative AI in programming education across Sub-Saharan Africa. *Computers and Education Open* 8 (2025). https://doi.org/10.1016/j.caeo.2025.100245

[168] Jay Rhald C. Padilla, Myron Darrel L. Montefalcon, and Alexander A. Hernandez. 2023. Language AI in Programming: A Case Study of ChatGPT in Higher Eduation Using Natural Language Processing. In *2023 IEEE 11th Conference on Systems, Process & Control (ICSPC)*. 276–281. https://doi.org/10.1109/ICSPC59664.2023.10420194

[169] Aadarsh Padiyath, Xinying Hou, Amy Pang, Diego Viramontes Vargas, Xingjian Gu, Tamara Nelson-Fromm, Zihan Wu, Mark Guzdial, and Barbara Ericson. 2024. Insights from Social Shaping Theory: The Appropriation of Large Language Models in an Undergraduate Programming Course. In *Proceedings of the 2024 ACM Conference on International Computing Education Research - Volume 1* (Melbourne, VIC, Australia) *(ICER '24)*. Association for Computing Machinery, New York, NY, USA, 114–130. https://doi.org/10.1145/3632620.3671098

[170] Matthew J Page, Joanne E McKenzie, et al. 2021. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 372 (2021). https://doi.org/10.1136/bmj.n71

[171] Daniel Palacios-Alonso, Jaime Urquiza-Fuentes, J. Ángel Velázquez-Iturbide, and Julio Guillén-García. 2024. Experiences and Proposals of Use of Generative AI in Advanced Software Courses. In *2024 IEEE Global Engineering Education Conference (EDUCON)*. 1–10. https://doi.org/10.1109/EDUCON60312.2024.10578869

[172] Paola Palomino-Flores, Ricardo Cristi-López, and David Paul. 2024. Mastering Ethical Horizons: Exploring AI Integration in Advanced Studies of Engineering,

Technology, and Informatics. In *2024 IEEE World Engineering Education Conference (EDUNINE)*. 1–6. https://doi.org/10.1109/EDUNINE60625.2024.10500654

[173] Ashley Pang and Frank Vahid. 2024. ChatGPT and Cheat Detection in CS1 Using a Program Autograding System. In *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1* (Milan, Italy) *(ITiCSE 2024)*. Association for Computing Machinery, New York, NY, USA, 367–373. https://doi.org/10.1145/3649217.3653558

[174] Maciej Pankiewicz and Ryan S. Baker. 2024. Navigating Compiler Errors with AI Assistance - A Study of GPT Hints in an Introductory Programming Course. In *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1* (Milan, Italy) *(ITiCSE 2024)*. Association for Computing Machinery, New York, NY, USA, 94–100. https://doi.org/10.1145/3649217.3653608

[175] Christos Papakostas, Christos Troussas, Akrivi Krouska, and Cleo Sgouropoulou. 2024. A Rule-Based Chatbot Offering Personalized Guidance in Computer Programming Education. In *Generative Intelligence and Intelligent Tutoring Systems: 20th International Conference, ITS 2024, Thessaloniki, Greece, June 10–13, 2024, Proceedings, Part II* (Thessaloniki, Greece). Springer-Verlag, Berlin, Heidelberg, 253–264. https://doi.org/10.1007/978-3-031-63031-6_22

[176] Jacob Penney, João Felipe Pimentel, Igor Steinmacher, and Marco A. Gerosa. 2023. Anticipating User Needs: Insights from Design Fiction on Conversational Agents for Computational Thinking. In *Chatbot Research and Design: 7th International Workshop, CONVERSATIONS 2023, Oslo, Norway, November 22–23, 2023, Revised Selected Papers* (Oslo, Norway). Springer-Verlag, Berlin, Heidelberg, 204–219. https://doi.org/10.1007/978-3-031-54975-5_12

[177] Manuela-Andreea Petrescu, Emilia-Loredana Pop, and Tudor-Dan Mihoc. 2023. Students' interest in knowledge acquisition in Artificial Intelligence. *Procedia Computer Science* 225 (2023), 1028–1036. https://doi.org/10.1016/j.procs.2023.10.090 27th International Conference on Knowledge Based and Intelligent Information and Engineering Sytems (KES 2023).

[178] Eric Poitras, Brent Crane, and Angela Siegel. 2024. Generative AI in Introductory Programming Instruction: Examining the Assistance Dilemma with LLM-Based Code Generators. In *Proceedings of the 2024 on ACM Virtual Global Computing Education Conference V. 1* (Virtual Event, NC, USA) *(SIGCSE Virtual 2024)*. Association for Computing Machinery, New York, NY, USA, 186–192. https://doi.org/10.1145/3649165.3690111

[179] Diana M. Popescu and David A. Joyner. 2024. ChatGPT's Performance on Problem Sets in an At-Scale Introductory Computer Science Course. In *Proceedings of the Eleventh ACM Conference on Learning @ Scale* (Atlanta, GA, USA) *(L@S '24)*. Association for Computing Machinery, New York, NY, USA, 486–490. https://doi.org/10.1145/3657604.3664701

[180] Diana M. Popescu and David A. Joyner. 2024. Novelty, Rigidity, and Complexity: Toward Developing AI-Resistant Assessments in an Introductory Computer Science Class. In *2024 IEEE Digital Education and MOOCS Conference (DEMOcon)*. 1–6. https://doi.org/10.1109/DEMOcon63027.2024.10748200

[181] Leo Porter and Daniel Zingaro. 2024. *Learn AI-Assisted Python Programming: With Github Copilot and ChatGPT*. Simon and Schuster.

[182] Seth Poulsen, Sami Sarsa, James Prather, Juho Leinonen, Brett A. Becker, Arto Hellas, Paul Denny, and Brent N. Reeves. 2024. Solving Proof Block Problems Using Large Language Models. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 1* (Portland, OR, USA) *(SIGCSE 2024)*. Association for Computing Machinery, New York, NY, USA, 1063–1069. https://doi.org/10.1145/3626252.3630928

[183] James Prather, Paul Denny, Juho Leinonen, Brett A. Becker, Ibrahim Albluwi, et al. 2023. The Robots Are Here: Navigating the Generative AI Revolution in Computing Education. In *Proceedings of the 2023 Working Group Reports on Innovation and Technology in Computer Science Education (ITiCSE-WGR '23)*. Association for Computing Machinery. https://doi.org/10.1145/3623762.3633499

[184] James Prather, Juho Leinonen, Natalie Kiesler, Jamie Gorson Benario, Sam Lau, Stephen MacNeil, Narges Norouzi, Simone Opel, Vee Pettit, Leo Porter, Brent N. Reeves, Jaromir Savelka, David H. Smith, Sven Strickroth, and Daniel Zingaro. 2025. Beyond the Hype: A Comprehensive Review of Current Trends in Generative AI Research, Teaching Practices, and Tools. In *2024 Working Group Reports on Innovation and Technology in Computer Science Education* (Milan, Italy) *(ITiCSE 2024)*. Association for Computing Machinery, New York, NY, USA, 300–338. https://doi.org/10.1145/3689187.3709614

[185] James Prather, Brent N. Reeves, Paul Denny, Brett A. Becker, Juho Leinonen, Andrew Luxton-Reilly, Garrett Powell, James Finnie-Ansley, and Eddie Antonio Santos. 2023. "It's Weird That it Knows What I Want": Usability and Interactions with Copilot for Novice Programmers. *ACM Trans. Comput.-Hum. Interact.* 31, 1, Article 4 (Nov. 2023), 31 pages. https://doi.org/10.1145/3617367

[186] James Prather, Brent N Reeves, Paul Denny, Juho Leinonen, Stephen MacNeil, Andrew Luxton-Reilly, João Orvalho, Amin Alipour, Ali Alfageeh, Thezyrie Amarouche, Bailey Kimmel, Jared Wright, Musa Blake, and Gweneth Barbre. 2025. Breaking the Programming Language Barrier: Multilingual Prompting to Empower Non-Native English Learners. In *Proceedings of the 27th Australasian Computing Education Conference (ACE '25)*. Association for Computing Machinery, New York, NY, USA, 74–84. https://doi.org/10.1145/3716640.3716649

[187] James Prather, Brent N Reeves, Juho Leinonen, Stephen MacNeil, Arisoa S Randrianasolo, Brett A. Becker, Bailey Kimmel, Jared Wright, and Ben Briggs. 2024. The Widening Gap: The Benefits and Harms of Generative AI for Novice Programmers. In *Proceedings of the 2024 ACM Conference on International Computing Education Research - Volume 1* (Melbourne, VIC, Australia) *(ICER '24)*. Association for Computing Machinery, New York, NY, USA, 469–486. https://doi.org/10.1145/3632620.3671116

[188] Cong Pu. 2024. Integrating Generative AI with Data Structures and Algorithm Analysis Course Homework. In *2024 IEEE Frontiers in Education Conference (FIE)*. 1–9. https://doi.org/10.1109/FIE61694.2024.10892822

[189] Shuying Qiao, Paul Denny, and Nasser Giacaman. 2025. Oversight in Action: Experiences with Instructor-Moderated LLM Responses in an Online Discussion Forum. In *Proceedings of the 27th Australasian Computing Education Conference (ACE '25)*. Association for Computing Machinery, New York, NY, USA, 95–104. https://doi.org/10.1145/3716640.3716651

[190] Basit Qureshi. 2023. ChatGPT in Computer Science Curriculum Assessment: An analysis of Its Successes and Shortcomings. In *Proceedings of the 2023 9th International Conference on E-Society, e-Learning and e-Technologies* (Portsmouth, United Kingdom) *(ICSLT '23)*. Association for Computing Machinery, New York, NY, USA, 7–13. https://doi.org/10.1145/3613944.3613946

[191] Gholamreza Rafiee, Firuza Ahmadli, and Matthew Collins. 2024. Fostering Personalized Learning in Data Science: Integrating Innovative Tools and Strategies for Diverse Pathways. In *2024 IEEE 13th International Conference on Engineering Education (ICEED)*. 1–6. https://doi.org/10.1109/ICEED62316.2024.10923798

[192] Parsa Rajabi, Parnian Taghipour, Diana Cukierman, and Tenzin Doleck. 2023. Exploring ChatGPT's impact on post-secondary education: A qualitative study. In *Proceedings of the 25th Western Canadian Conference on Computing Education* (Vancouver, BC, Canada) *(WCCCE '23)*. Association for Computing Machinery, New York, NY, USA, Article 9, 6 pages. https://doi.org/10.1145/3593342.3593360

[193] Jaakko Rajala, Jenni Hukkanen, Maria Hartikainen, and Pia Niemel¨a. 2023. """¨Call me Kiran¨ – ChatGPT as a Tutoring Chatbot in a Computer Science Course"". In *Proceedings of the 26th International Academic Mindtrek Conference* (Tampere, Finland) *(Mindtrek '23)*. Association for Computing Machinery, New York, NY, USA, 83–94. https://doi.org/10.1145/3616961.3616974

[194] Tlou RAMABU and Tumelo MALEBANE. 2024. Guidelines for Effective Use of ChatGPT in Introductory Programming Education. In *2024 IST-Africa Conference (IST-Africa)*. 1–8. https://doi.org/10.23919/IST-Africa63983.2024.10569684

[195] Vijayalakshmi Ramasamy, Suganya Ramamoorthy, Gursimran Singh Walia, Eli Kulpinski, and Aaron Antreassian. 2024. Enhancing User Story Generation in Agile Software Development Through Open AI and Prompt Engineering. In *2024 IEEE Frontiers in Education Conference (FIE)*. 1–8. https://doi.org/10.1109/FIE61694.2024.10893343

[196] Aninditha Ramesh, Arav Agarwal, Jacob Arthur Doughty, Ketan Ramaneti, Jaromir Savelka, and Majd Sakr. 2024. A Benchmark for Testing the Capabilities of LLMs in Assessing the Quality of Multiple-choice Questions in Introductory Programming Education. In *Proceedings of the 2024 on ACM Virtual Global Computing Education Conference V. 1* (Virtual Event, NC, USA) *(SIGCSE Virtual 2024)*. Association for Computing Machinery, New York, NY, USA, 193–199. https://doi.org/10.1145/3649165.3690123

[197] Valeria Ramirez Osorio, Angela Zavaleta Bernuy, Bogdan Simion, and Michael Liut. 2025. Understanding the Impact of Using Generative AI Tools in a Database Course. In *Proceedings of the 56th ACM Technical Symposium on Computer Science Education V. 1* (Pittsburgh, PA, USA) *(SIGCSETS 2025)*. Association for Computing Machinery, New York, NY, USA, 959–965. https://doi.org/10.1145/3641554.3701785

[198] Sanka Rasnayaka, Guanlin Wang, Ridwan Shariffdeen, and Ganesh Neelakanta Iyer. 2024. An Empirical Study on Usage and Perceptions of LLMs in a Software Engineering Project. In *Proceedings of the 1st International Workshop on Large Language Models for Code* (Lisbon, Portugal) *(LLM4Code '24)*. Association for Computing Machinery, New York, NY, USA, 111–118. https://doi.org/10.1145/3643795.3648379

[199] Brent Reeves, Sami Sarsa, James Prather, Paul Denny, Brett A. Becker, Arto Hellas, Bailey Kimmel, Garrett Powell, and Juho Leinonen. 2023. Evaluating the Performance of Code Generation Models for Solving Parsons Problems With Small Prompt Variations. In *Proceedings of the 2023 Conference on Innovation and Technology in Computer Science Education V. 1 (ITiCSE 2023)*. ACM.

[200] Brent N. Reeves, James Prather, Paul Denny, Juho Leinonen, Stephen MacNeil, Brett A. Becker, and Andrew Luxton-Reilly. 2024. Prompts First, Finally. arXiv:2407.09231 [cs.CY] https://arxiv.org/abs/2407.09231

[201] Junghwan Rhee, Aakankshya Shrestha, Gang Qian, Fei Zuo, Jicheng Fu, Myungah Park, Xianshan Qu, Goutam Mylavarapu, and Hong Sung. 2024. An Evaluation on the Impact of Large Language Models on Computer Science Curricula. *Journal of Computing Sciences in Colleges* 40, 1 (2024), 70–80. https://doi.org/10.5555/3715602.3715614

[202] Mike Richards, Kevin Waugh, Mark Slaymaker, Marian Petre, John Woodthorpe, and Daniel Gooch. 2024. Bob or Bot: Exploring ChatGPT's Answers to University Computer Science Assessment. *ACM Trans. Comput. Educ.* 24, 1, Article 5 (Jan. 2024), 32 pages. https://doi.org/10.1145/3633287

[203] Elijah Rivera, Alexander Steinmaurer, Kathi Fisler, and Shriram Krishnamurthi. 2024. Iterative Student Program Planning using Transformer-Driven Feedback. In *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1* (Milan, Italy) *(ITiCSE 2024)*. Association for Computing Machinery, New York, NY, USA, 45–51. https://doi.org/10.1145/3649217.3653607

[204] Lianne Roest, Hieke Keuning, and Johan Jeuring. 2024. Next-step hint generation for introductory programming using large language models. In *Proceedings of the 26th Australasian Computing Education Conference*. 144–153.

[205] Lianne Roest, Hieke Keuning, and Johan Jeuring. 2024. Next-Step Hint Generation for Introductory Programming Using Large Language Models. In *Proceedings of the 26th Australasian Computing Education Conference* (Sydney, NSW, Australia) *(ACE '24)*. Association for Computing Machinery, New York, NY, USA, 144–153. https://doi.org/10.1145/3636243.3636259

[206] Kantwon Rogers, Michael Davis, Mallesh Maharana, Pete Etheredge, and Sonia Chernova. 2025. Playing Dumb to Get Smart: Creating and Evaluating an LLM-based Teachable Agent within University Computer Science Classes. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, Article 126, 22 pages. https://doi.org/10.1145/3706598.3713644

[207] Michael P. Rogers, Hannah Miller Hillberg, and Christopher L. Groves. 2024. Attitudes Towards the Use (and Misuse) of ChatGPT: A Preliminary Study. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 1* (Portland, OR, USA) *(SIGCSE 2024)*. Association for Computing Machinery, New York, NY, USA, 1147–1153. https://doi.org/10.1145/3626252.3630784

[208] Michael P. Rogers, Hannah Miller Hillberg, and Christopher L. Groves. 2024. Attitudes Towards the Use (and Misuse) of ChatGPT: A Preliminary Study. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 1* (Portland, OR, USA) *(SIGCSE 2024)*. Association for Computing Machinery, New York, NY, USA, 1147–1153. https://doi.org/10.1145/3626252.3630784

[209] Smitha S Kumar, Michael Adam Lones, Manuel Maarek, and Hind Zantout. 2024. Investigating the Proficiency of Large Language Models in Formative Feedback Generation for Student Programmers. In *Proceedings of the 1st International Workshop on Large Language Models for Code* (Lisbon, Portugal) *(LLM4Code '24)*. Association for Computing Machinery, New York, NY, USA, 88–93. https://doi.org/10.1145/3643795.3648380

[210] Saiful Islam Salim, Rubin Yuchan Yang, Alexander Cooper, Suryashree Ray, Saumya Debray, and Sazzadur Rahaman. 2024. Impeding LLM-assisted Cheating in Introductory Programming Assignments via Adversarial Perturbation. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (Eds.). Association for Computational Linguistics, Miami, Florida, USA, 445–463. https://doi.org/10.18653/v1/2024.emnlp-main.27

[211] Rajagopal Sankaranarayanan and Shamima Mithun. 2024. Exploring the Effectiveness of AI-Enabled Microlearning in Database Design and Programming Course. In *2024 IEEE Frontiers in Education Conference (FIE)*. 1–7. https://doi.org/10.1109/FIE61694.2024.10892916

[212] Thulasi M. Santhi and K. Srinivasan. 2024. ChatGPT-Based Learning Platform for Creation of Different Attack Model Signatures and Development of Defense Algorithm for Cyberattack Detection. *IEEE Transactions on Learning Technologies* 17 (2024), 1829–1842. https://doi.org/10.1109/TLT.2024.3417252

[213] Eddie Antonio Santos and Brett A. Becker. 2024. Not the Silver Bullet: LLM-enhanced Programming Error Messages are Ineffective in Practice. In *Proceedings of the 2024 Conference on United Kingdom & Ireland Computing Education Research* (Manchester, United Kingdom) *(UKICER '24)*. Association for Computing Machinery, New York, NY, USA, Article 5, 7 pages. https://doi.org/10.1145/3689535.3689554

[214] Sami Sarsa, Paul Denny, Arto Hellas, and Juho Leinonen. 2022. Automatic Generation of Programming Exercises and Code Explanations Using Large Language Models. In *Proceedings of the 2022 ACM Conference on International Computing Education Research - Volume 1* (Lugano and Virtual Event, Switzerland) *(ICER '22)*. Association for Computing Machinery, New York, NY, USA, 27–43. https://doi.org/10.1145/3501385.3543957

[215] Sami Sarsa, Paul Denny, Arto Hellas, and Juho Leinonen. 2022. Automatic Generation of Programming Exercises and Code Explanations Using Large Language Models. In *Proceedings of the 2022 ACM Conference on International Computing Education Research - Volume 1* (Lugano and Virtual Event, Switzerland) *(ICER '22)*. Association for Computing Machinery, New York, NY, USA, 27–43. https://doi.org/10.1145/3501385.3543957

[216] Fatemeh Sarshartehrani, Elham Mohammadrezaei, Majid Behravan, and Denis Gracanin. 2024. Enhancing e-learning experience through embodied ai tutors in immersive virtual environments: A multifaceted approach for personalized educational adaptation. In *International Conference on Human-Computer Interaction*. Springer, 272–287. https://doi.org/10.1007/978-3-031-60609-0_20

[217] Jaromir Savelka, Arav Agarwal, Marshall An, Chris Bogart, and Majd Sakr. 2023. Thrilled by Your Progress! Large Language Models (GPT-4) No Longer Struggle to Pass Assessments in Higher Education Programming Courses. In *Proceedings of the 2023 ACM Conference on International Computing Education Research*. ACM. https://doi.org/10.1145/3568813.3600142

[218] Jaromir Savelka, Arav Agarwal, Christopher Bogart, Yifan Song, and Majd Sakr. 2023. Can generative pre-trained transformers (gpt) pass assessments in higher education programming courses?. In *Proceedings of the 2023 Conference on Innovation and Technology in Computer Science Education V. 1*. 117–123. https://doi.org/10.1145/3587102.3588792

[219] Sigrid Schefer-Wenzl, Christoph Vogl, Sahani Peiris, and Igor Miladinovic. 2024. Exploring the adoption of generative AI tools in computer science education: A student survey. In *Proceedings of the 2024 the 16th International Conference on Education Technology and Computers*. 173–178. https://doi.org/10.1145/3702163.3702188

[220] Andreas Scholl and Natalie Kiesler. 2024. How Novice Programmers Use and Experience ChatGPT when Solving Programming Exercises in an Introductory Course. In *2024 IEEE Frontiers in Education Conference (FIE)*. IEEE, 1–9. https://doi.org/10.1109/FIE61694.2024.10893442

[221] Andreas Scholl, Daniel Schiffner, and Natalie Kiesler. 2024. Analyzing Chat Protocols of Novice Programmers Solving Introductory Programming Tasks with ChatGPT. *arXiv preprint arXiv:2405.19132* (2024).

[222] Christian Servin, Nadia V. Karichev, and Myshie Pagel. 2024. Unfolding Programming: How to Use AI Tools in Introductory Computing Courses. In *Proceedings of the 25th Annual Conference on Information Technology Education (SIGITE 2024)*. 49–55. https://doi.org/10.1145/3686852.3687073

[223] Judy Sheard, Paul Denny, Arto Hellas, Juho Leinonen, Lauri Malmi, and Simon. 2024. Instructor Perceptions of AI Code Generation Tools - A Multi-Institutional Interview Study. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 1 (SIGCSE 2024)*. ACM, 7 pages. https://doi.org/10.1145/3626252.3630880

[224] Brad Sheese, Mark Liffiton, Jaromir Savelka, and Paul Denny. 2024. Patterns of Student Help-Seeking When Using a Large Language Model-Powered Programming Assistant. In *Proceedings of the 26th Australasian Computing Education Conference* (Sydney, NSW, Australia) *(ACE '24)*. Association for Computing Machinery, New York, NY, USA, 49–57. https://doi.org/10.1145/3636243.3636249

[225] Yiyin Shen, Xinyi Ai, Adalbert Gerald Soosai Raj, and Meenakshi Syamkumar. 2024. Implications of ChatGPT for Data Science Education. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education (SIGCSE '24)* (Portland, OR, USA). ACM, 1230–1236. https://doi.org/10.1145/3626252.3630874

[226] Mario Simaremare, Chandro Pardede, Irma Tampubolon, Putri Manurung, and Daniel Simangunsong. 2024. Pair Programming in Programming Courses in the Era of Generative AI: Students' Perspective. In *2024 31st Asia-Pacific Software Engineering Conference (APSEC)*. 507–511. https://doi.org/10.1109/APSEC65559.2024.00069

[227] Harpreet Singh, Mohammad-Hassan Tayarani-Najaran, and Muhammad Yaqoob. 2023. Exploring computer science students' perception of ChatGPT in higher education: A descriptive and correlation study. *Education Sciences* 13, 9 (2023), 924. https://doi.org/10.3390/educsci13090924

[228] C. Estelle Smith, Kylee Shiekh, Hayden Cooreman, Sharfi Rahman, Yifei Zhu, Md Kamrul Siam, Michael Ivanitskiy, Ahmed M. Ahmed, Michael Hallinan, Alexander Grisak, and Gabe Fierro. 2024. Early Adoption of Generative Artificial Intelligence in Computing Education: Emergent Student Use Cases and Perspectives in 2023. In *Proceedings of the 2024 Innovation and Technology in Computer Science Education V.1 (ITiCSE 2024)* (Milan, Italy). ACM, 3–9. https://doi.org/10.1145/3649217.3653575

[229] David H. IV Smith, Max Fowler, and Paul Denny. 2024. Prompting for Comprehension: Exploring the Intersection of Explain in Plain English Questions and Prompt Writing. In *Proceedings of the 11th ACM Conference on Learning@Scale (L@S '24)*. ACM, 39–50. https://doi.org/10.1145/3657604.3662039

[230] David Soto, Manabu Higashida, Shizuka Shirai, Mayumi Ueda, and Yuki Uranishi. 2024. Enhancing Learning Dynamics: Integrating Interactive Learning Environments and ChatGPT for Computer Networking Lessons. *Procedia Computer Science* 227 (2024), 1399–1408. https://doi.org/10.1016/j.procs.2024.05.207

[231] Aiste Steponenaite and Basel Barakat. 2023. Plagiarism in AI Empowered World. In *Universal Access in Human-Computer Interaction: 17th International Conference, UAHCI 2023, Held as Part of the 25th HCI International Conference, HCII 2023, Copenhagen, Denmark, July 23–28, 2023, Proceedings, Part II* (Copenhagen, Denmark). Springer-Verlag, Berlin, Heidelberg, 434–442. https://doi.org/10.1007/978-3-031-35897-5_31

[232] Anselm L Strauss and Juliet Corbin. 2004. Open coding. *Social research methods: A reader* (2004), 303–306.

[233] Arne Styve, Outi T Virkki, and Usman Naeem. 2024. Developing critical thinking practices interwoven with generative AI usage in an introductory programming course. In *2024 IEEE Global Engineering Education Conference (EDUCON)*. IEEE, 01–08. https://doi.org/10.1109/EDUCON60312.2024.10578746

[234] Luhang Sun, Mian Wei, Yibing Sun, Yoo Ji Suh, Liwei Shen, and Sijia Yang. 2024. Smiling women pitching down: auditing representational and presentational gender biases in image-generative AI. *Journal of Computer-Mediated Communication* 29, 1 (02 2024), zmad045. https://doi.org/10.1093/jcmc/zmad045 arXiv:https://academic.oup.com/jcmc/article-pdf/29/1/zmad045/56546560/zmad045.pdf

[235] Wannita Takerngsaksiri, Cleshan Warusavitarne, Christian Yaacoub, Matthew Hee Keng Hou, and Chakkrit Tantithamthavorn. 2024. Students' Perspectives on AI Code Completion: Benefits and Challenges. In *2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)*. 1606–1611. https://doi.org/10.1109/COMPSAC61105.2024.00252

[236] Ben Arie Tanay, Lexy Arinze, Siddhant S Joshi, Kirsten A Davis, and James C Davis. 2024. An Exploratory Study on Upper-Level Computing Students' Use of Large Language Models as Tools in a Semester-Long Project. *2024 ASEE Annual Conference & Exposition* (2024). https://doi.org/10.18260/1-2--46557

[237] Xiaohang Tang, Sam Wong, Marcus Huynh, Zicheng He, Yalong Yang, and Yan Chen. 2025. SPHERE: Supporting Personalized Feedback at Scale in Programming Classrooms with Structured Review of Generative AI Outputs. In *Proceedings of the 2025 ACM CHI Conference on Human Factors in Computing Systems (CHI EA '25)*. https://doi.org/10.1145/3706599.3720203

[238] Andrew Taylor, Alexandra Vassar, Jake Renzella, and Hammond Pearce. 2024. dcc –help: Transforming the Role of the Compiler by Generating Context-Aware Error Explanations with Large Language Models. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V. 1* (Portland, OR, USA) *(SIGCSE 2024)*. Association for Computing Machinery, New York, NY, USA, 1314–1320. https://doi.org/10.1145/3626252.3630822

[239] Vincent Tinto. 1975. Dropout from higher education: A theoretical synthesis of recent research. *Review of educational research* 45, 1 (1975), 89–125. https://doi.org/10.3102/00346543045001089

[240] Christos Troussas, Christos Papakostas, Akrivi Krouska, Phivos Mylonas, and Cleo Sgouropoulou. 2024. Evaluating ChatGPT-driven Automated Test Generation for Personalized Programming Education. In *2024 2nd International Conference on Foundation and Large Language Models (FLLM)*. 194–200. https://doi.org/10.1109/FLLM63129.2024.10852510

[241] Chia-Wen Tsai, Lynne Lee, Michael Yu-Ching Lin, Yih-Ping Cheng, Chih-Hsien Lin, and Meng-Chuan Tsai. 2025. Effects of integrating self-regulation scaffolding supported by chatbot and online collaborative reflection on students' learning in an artificial intelligence course. *Computers & Education* 232 (2025), 105305. https://doi.org/10.1016/j.compedu.2025.105305

[242] Ayca Tuzmen. 2024. Use of Generative Artificial Intelligence in the Education of Software Verification and Validation. In *2024 IEEE Frontiers in Education Conference (FIE)*. 1–6. https://doi.org/10.1109/FIE61694.2024.10893333

[243] Annapurna Vadaparty, Daniel Zingaro, David H. Smith IV, Mounika Padala, Christine Alvarado, Jamie Gorson Benario, and Leo Porter. 2024. CS1-LLM: Integrating LLMs into CS1 Instruction. In *Proceedings of the 2024 on Innovation and Technology in Computer Science Education V. 1* (Milan, Italy) *(ITiCSE 2024)*. Association for Computing Machinery, New York, NY, USA, 297–303. https://doi.org/10.1145/3649217.3653584

[244] Kevin Wang, Seth Akins, Abdallah Mohammed, and Ramon Lawrence. 2023. Student Mastery or AI Deception? Analyzing ChatGPT's Assessment Proficiency and Evaluating Detection Strategies. In *2023 International Conference on Computational Science and Computational Intelligence (CSCI)*. 1615–1621. https://doi.org/10.1109/CSCI62032.2023.00268

[245] Tianjia Wang, Daniel Vargas Díaz, Chris Brown, and Yan Chen. 2023. Exploring the Role of AI Assistants in Computer Science Education: Methods, Implications, and Instructor Perspectives. In *2023 IEEE Symposium on Visual Languages and Human-Centric Computing (VL/HCC)*. 92–102. https://doi.org/10.1109/VL-HCC57772.2023.00018

[246] Yang Wang, Margaret McCoey, Qian Hu, and Maryam Jalalitabar. 2024. Teaching Security in the Era of Generative AI: A Course Design of Security + ChatGPT. In *2024 IEEE Integrated STEM Education Conference (ISEC)*. 01–06. https://doi.org/10.1109/ISEC61299.2024.10664905

[247] Michel Wermelinger. 2023. Using GitHub Copilot to Solve Simple Programming Problems. In *Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 1* (Toronto ON, Canada) *(SIGCSE 2023)*. Association for Computing Machinery, New York, NY, USA, 172–178. https://doi.org/10.1145/3545945.3569830

[248] Jan H.R. Woerner, Aleksandra P. Turtova, and Andrew S.I.D. Lang. 2024. Transformative Potentials and Ethical Considerations of AI Tools in Higher Education: Case Studies and Reflections. In *SoutheastCon 2024*. 510–515. https://doi.org/10.1109/SoutheastCon52093.2024.10500042

[249] Ruiwei Xiao, Xinying Hou, Harsh Kumar, Steven Moore, John Stamper, and Michael Liut. 2024. A Preliminary Analysis of Students' Help Requests with an LLM-powered Chatbot when Completing CS1 Assignments. CSEDM'24: 8th Educational Data Mining in Computer Science Education (CSEDM).

[250] Yuankai Xue, Hanlin Chen, Gina R. Bai, Robert Tairas, and Yu Huang. 2024. Does ChatGPT Help With Introductory Programming?An Experiment of Students Using ChatGPT in CS1. In *Proceedings of the 46th International Conference on Software Engineering: Software Engineering Education and Training* (Lisbon, Portugal) *(ICSE-SEET '24)*. Association for Computing Machinery, New York, NY, USA, 331–341. https://doi.org/10.1145/3639474.3640076

[251] Thomas Y. Yeh, Karena Tran, Ge Gao, Tyler Yu, Wai On Fong, and Tzu-Yi Chen. 2025. Bridging Novice Programmers and LLMs with Interactivity. In *Proceedings of the 56th ACM Technical Symposium on Computer Science Education V. 1* (Pittsburgh, PA, USA) *(SIGCSETS 2025)*. Association for Computing Machinery, New York, NY, USA, 1295–1301. https://doi.org/10.1145/3641554.3701867

[252] Zezhu Yu, Suqing Liu, Paul Denny, Andreas Bergen, and Michael Liut. 2025. Integrating Small Language Models with Retrieval-Augmented Generation in Computing Education: Key Takeaways, Setup, and Practical Insights. In *Proceedings of the 56th ACM Technical Symposium on Computer Science Education V. 1* (Pittsburgh, PA, USA) *(SIGCSETS 2025)*. Association for Computing Machinery, New York, NY, USA, 1302–1308. https://doi.org/10.1145/3641554.3701844

[253] J.D. Zamfirescu-Pereira, Laryn Qi, Bjorn Hartmann, John DeNero, and Narges Norouzi. 2025. 61A Bot Report: AI Assistants in CS1 Save Students Homework Time and Reduce Demands on Staff. (Now What?). In *Proceedings of the 56th ACM Technical Symposium on Computer Science Education V. 1* (Pittsburgh, PA, USA) *(SIGCSETS 2025)*. Association for Computing Machinery, New York, NY, USA, 1309–1315. https://doi.org/10.1145/3641554.3701864

[254] Cynthia Zastudil, Magdalena Rogalska, Christine Kapp, Jennifer Vaughn, and Stephen MacNeil. 2023. Generative AI in Computing Education: Perspectives of Students and Instructors. In *2023 IEEE Frontiers in Education Conference (FIE)*. https://doi.org/10.1109/FIE58773.2023.10343467

[255] Katerina Zdravkova, Fisnik Dalipi, Fredrik Ahlgren, Bojan Ilijoski, and Tobias Olsson. 2024. Unveiling the impact of large language models on student learning: A comprehensive case study. In *2024 IEEE Global Engineering Education Conference (EDUCON)*. 1–8. https://doi.org/10.1109/EDUCON60312.2024.10578855

[256] Jialu Zhang, José Pablo Cambronero, Sumit Gulwani, Vu Le, Ruzica Piskac, Gustavo Soares, and Gust Verbruggen. 2024. PyDex: Repairing Bugs in Introductory Python Assignments using LLMs. *Proc. ACM Program. Lang.* 8, OOPSLA1, Article 133 (April 2024), 25 pages. https://doi.org/10.1145/3649850

[257] Fei Zuo, Cody Tompkins, Gang Qian, Junghwan Rhee, Xianshan Qu, and Bokai Yang. 2024. ChatGPT as an Assembly Language Interpreter for Computing Education. *J. Comput. Sci. Coll.* 40, 2 (Oct. 2024), 73–82. https://doi.org/10.5555/3715622.3715633

[258] Rina Ziel-Girshin. 2024. The Good and Bad of AI Tools in Novice Programming Education. *Education Sciences* 14, 10 (2024), 1089. https://doi.org/10.3390/educsci14101089